Biology Grade 11 - Solved Exercises (BTBB)

Balochistan Textbook Board (BTBB)

Chapter 1: Cell Structure and Function

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. What structure surrounds all cells and controls what enters and leaves?

- Cytoplasm
- ✓ Cell membrane
- Cell wall
- Nucleus

Q2. Identify the organelle present only in animal cells.

- Chloroplast
- Nucleus
- ✓ Centriole
- Ribosome

Q3. All of the following contain ribosomes except:

- Mitochondria
- Chloroplast
- ✓ Golgi apparatus
- Nucleus

Q4. The fluid mosaic model describes the cell membrane as:

- A rigid and immoveable barrier
- Fixed protein layer
- ✓ Flexible layer with moving parts

A crystalline structure

Q5. The cell organelle that has DNA.

✓ Mitochondrion

Golgi body

Lysosome

Endoplasmic reticulum

Q6. Lysosomes are called suicidal bags because:

Contain hydrolytic enzymes

Remove waste from cell

Contain food vacuole

✓ Destroy its own cell

Q7. The matrix of chloroplast is called:

Granum

✓ Stroma

Lamella

Plasma

Q8. What does the RER have that the SER does not?

Lipids

Lysosome

Nucleus

✓ Ribosomes

Q9. The mosaic nature of cell membrane is due to:

Lipid

✓ Protein

Carbohydrate

Steroid

Q10. Cell inclusions are:

✓ Non living materials present in cytoplasm

Another name of cell organelles

Cytoskeleton framework of cell

Combined name for cell wall and plasma membrane

Q11. Which of these is wrongly matched?

Chromoplast-Pollination

Chloroplast-Chlorophyll

Golgi bodies-Secretion

✓ Ribosome-Powerhouse

Q12. The outermost covering of animal cells is:

Cell Wall

✓ Cell Membrane

Nuclear Membrane

Ribosome

Q13. Which part of nucleus makes ribosomal sub units?

Nuclear membrane

✓ Nucleolus

Nucleoplasm

Chromatin

Q14. In many animal cells the cell membrane helps to take in materials by infolding. This type of intake is termed as:

- ✓ Endocytosis
- Phagocytosis
- Pinocytosis
- Exocytosis

Q15. The breakdown and formation of hydrogen peroxide takes place in:

- Ribosome
- Lysosome
- Golgi body
- ✓ Peroxisome

Q16. The movement of molecules from higher to lower concentration with specific carrier protein is called:

- Simple diffusion
- ✓ Facilitated Diffusion
- Endosomes and Exosomes
- Passive transport

Q17. Which of the following is used in electron microscope?

- Electron beams
- Magnetic field
- Light waves
- ✓ Electron beams and magnetic field

Q18. Which of the following is best suited to get the 3D surface

view of an object?

✓ SEM

TEM

Both a and b

Compound microscope

Q19. Cell signaling is:

Intercellular

Intra cellular

✓ Both a and b

None of the above

Q20. Microfilaments are mostly made up of:

Tubulin

Keratin

Tropomyosin

✓ Actin

Exercise - Section II: Short Response Questions

Q1. Why is mitochondria called the powerhouse of the cell?

Ans. Mitochondria are called the powerhouse of the cell because they are the sites of cellular respiration. They oxidise food molecules and release the energy stored in them in the form of ATP, the energy currency which powers almost all activities of the cell.

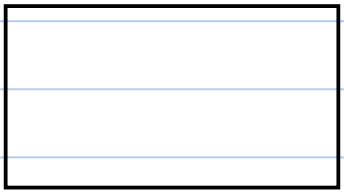
Q2. How are mitochondria similar to bacteria?

Ans. Mitochondria resemble bacteria in having their own circular

DNA, their own ribosomes (70S type) and a double membrane, and in dividing by a fission-like process. These similarities support the idea that mitochondria evolved from free-living bacteria that came to live inside early eukaryotic cells.

Q3. Draw a labeled diagram of a chloroplast.

Ans. A chloroplast is bounded by a double membrane (outer and inner membrane). Inside is the fluid stroma, in which stacks of disc-like thylakoids called grana are present; the grana are connected by stroma lamellae. The diagram below shows these parts labeled: outer membrane, inner membrane, stroma, granum (thylakoid stack) and lamella.



Chloroplast: outer membrane, inner membrane, stroma, granum (stack of thylakoids), stroma lamella

Q4. Which organelle is called the factory of protein synthesis and why?

Ans. The ribosome is called the factory of protein synthesis because proteins are actually assembled on it. The ribosome reads the message of mRNA and joins amino acids together in the correct order to form a polypeptide chain, just as a factory assembles a product.

Q5. How would you differentiate between a prokaryotic cell and a eukaryotic cell?

Ans. A prokaryotic cell has no true nucleus or membrane-bound organelles; its single circular chromosome lies free in the cytoplasm and it divides by binary fission. A eukaryotic cell has a true nucleus bounded by a nuclear membrane, many membrane-bound organelles, two or more linear chromosomes, and it divides by mitosis or meiosis.

Q6. Why is the lysosome known as a digesting organelle and suicidal bag?

Ans. The lysosome is a digesting organelle because it contains about forty hydrolytic enzymes that digest food particles, worn-out organelles and foreign material taken into the cell. It is called a suicidal bag because if its membrane bursts, these enzymes digest the cell's own contents and the cell dies.

Q7. What is the role of lysosome in crinophagy?

Ans. In crinophagy, excess secretory granules (for example unused hormone vesicles) fuse with lysosomes. The lysosomal enzymes then digest the contents of the granules, so the cell safely disposes of secretory material it no longer needs instead of releasing it.

Q8. How would you differentiate between: (a) Prokaryotic cell wall and eukaryotic cell wall (b) Primary lysosome and secondary lysosome (c) Resolution and magnification power of microscope (d) RER and SER (e) Endocytosis and exocytosis?

Ans. (a) The prokaryotic (bacterial) wall is made of peptidoglycan, while the eukaryotic plant wall is made of cellulose and the fungal wall of chitin; animal cells have no wall. (b) A primary lysosome is a newly formed vesicle containing inactive enzymes, while a secondary lysosome is formed when a primary lysosome fuses with a food vacuole and digestion is in progress. (c) Resolution is the ability to distinguish two close objects as separate, while magnification is only the increase in apparent size. (d) RER bears ribosomes on its surface and makes proteins, while SER has no ribosomes and makes lipids and detoxifies. (e) Endocytosis takes material into the cell by infolding of the membrane, while exocytosis expels material from the cell by fusion of vesicles with the membrane.

Q9. Differences between glyoxisome and peroxisome.

Ans. Both are microbodies, but glyoxisomes occur only in plants (mainly in germinating seeds) and convert stored fats into carbohydrates through the glyoxylate cycle. Peroxisomes occur in both plant and animal cells and mainly break down hydrogen peroxide and carry out oxidative reactions such as the breakdown of fatty acids.

Q10. What is the role of nucleolus?

Ans. The nucleolus is the factory of ribosomes: it synthesises ribosomal RNA and assembles it with proteins to form the small and large ribosomal subunits. These subunits then pass through the nuclear pores into the cytoplasm, where they join to form working ribosomes.

Q13. How does lysosome differ from glyoxisome?

Ans. A lysosome is a membrane-bound sac of hydrolytic enzymes that digests food, foreign particles and worn-out cell parts in animal and plant cells. A glyoxisome is a microbody found only in plants which contains enzymes of the glyoxylate cycle and converts fats into sugars in germinating seeds; it does not carry out general intracellular digestion.

Q14. How are mitochondria and chloroplast similar to a prokaryotic cell?

Ans. Mitochondria and chloroplasts both have their own circular DNA, 70S ribosomes and a double membrane, and they divide by fission, like bacteria. This shows they probably originated as free-living prokaryotes that began living inside early eukaryotic cells.

Q15. Who proposed the fluid mosaic model of plasma membrane?

Ans. The fluid mosaic model of the plasma membrane was proposed by S. J. Singer and G. L. Nicolson in 1972. It describes the membrane as a fluid lipid bilayer in which proteins float like icebergs, giving it a mosaic appearance.

Q16. Difference between cisternae and tubules.

Ans. Cisternae are flattened, sac-like compartments, for example the flattened sacs of the Golgi apparatus and rough endoplasmic reticulum. Tubules are narrow tube-like channels, for example the tubular network of the smooth endoplasmic reticulum; both are

parts of the endomembrane system but differ in shape.

Q17. What is ergastoplasm?

Ans. Ergastoplasm is the older name for the rough endoplasmic reticulum, the cytoplasm rich in ribosome-studded membranes seen in actively protein-secreting cells. Because of its many ribosomes it stains strongly with basic dyes, and it is the site where proteins for export are synthesised.

Q18. Define protein signaling.

Ans. Protein signaling (the fixed membrane pathway) is the cell-signaling pathway used by water-soluble peptide and protein hormones that cannot cross the plasma membrane. The hormone (first messenger) binds a receptor on the cell surface, which activates adenylate cyclase to produce cAMP (the second messenger); cAMP then triggers changes inside the cell such as enzyme activation and glycogen breakdown.

Q19. How does protein signaling differ from steroid signaling?

Ans. In protein signaling the hormone is water-soluble, cannot enter the cell, binds a surface receptor and acts through a second messenger (cAMP) without changing gene expression directly. In steroid signaling the hormone is lipid-soluble, passes through the membrane, binds an intracellular receptor, and the hormone-receptor complex acts as a transcription factor that changes gene expression and produces new proteins.

Q20. Why do animal cells not possess a cell wall?

Ans. Animal cells do not need a rigid wall because they do not take up water by osmosis to the point of bursting in the same way, and they need a flexible surface for movement, change of shape and processes like phagocytosis. Support and shape are instead provided by the cytoskeleton, and the plasma membrane alone forms the outer boundary.

Exercise - Section III: Extended Response Questions

Q1. Explain the structure of the plant cell wall.

Ans. The plant cell wall is a rigid outer layer outside the plasma membrane. Its primary wall is made mainly of cellulose microfibrils embedded in pectin and hemicellulose, which gives strength with flexibility for growth. Many mature cells add a thicker secondary wall rich in cellulose and lignin. Between neighbouring cells lies the middle lamella of calcium pectate, and narrow plasmodesmata channels connect the cytoplasm of adjacent cells.

Q2. Explain the fluid mosaic model of plasma membrane. Write down some important functions of plasma membrane.

Ans. According to the Singer-Nicolson fluid mosaic model (1972), the plasma membrane is a bilayer of phospholipids in which proteins are embedded like a mosaic and can move sideways, so the membrane behaves like a fluid. Cholesterol, glycoproteins and glycolipids are also present. Its functions are: it protects the cell and gives it shape; it controls what enters and leaves (selective permeability); it carries out transport by diffusion, facilitated diffusion and active transport; and its receptors receive signals and recognise other cells.

Q3. Explain the structure of ribosome and Golgi complex.

Ans. A ribosome is a small, non-membranous organelle made of rRNA and protein, with a large and a small subunit; in eukaryotes it is 80S (60S + 40S) and it may float free in the cytoplasm or sit on the RER. The Golgi complex is a stack of flattened membrane sacs (cisternae) with a forming (cis) face and a maturing (trans) face, plus associated vesicles. It modifies, packages and secretes proteins and lipids received from the ER, and forms lysosomes and cell-wall material.

Q4. Describe the structure of mitochondria and its role in cellular respiration.

Ans. A mitochondrion is bounded by a smooth outer membrane and a folded inner membrane whose folds are called cristae; the inner fluid is the matrix, which contains its own DNA, ribosomes and enzymes. In cellular respiration, pyruvate is broken down in the matrix (Krebs cycle) and the electron transport chain on the cristae releases energy to form ATP. Because it supplies most of the cell's ATP, the mitochondrion is called the powerhouse of the cell.

Q5. Illustrate the structure of the nucleus.

Ans. The nucleus is a large, usually spherical organelle bounded by a double nuclear membrane perforated by nuclear pores that control exchange with the cytoplasm. Inside is the nucleoplasm containing chromatin (DNA + protein), which condenses into chromosomes during division, and one or more nucleoli that make ribosomal subunits. The nucleus controls all cell activities by

directing protein synthesis through DNA, and it passes hereditary information to daughter cells.

Q6. Describe the role of chloroplast in photosynthesis.

Ans. The chloroplast is the site of photosynthesis in plant cells. Its pigment chlorophyll, present in the thylakoid membranes of the grana, absorbs light energy, which drives the light reactions that split water and produce ATP and NADPH. In the stroma, the dark reactions (Calvin cycle) use ATP and NADPH to fix carbon dioxide into sugars. Thus the chloroplast converts light energy into chemical energy stored in food.

Q7. Explain the fluid mosaic model of plasma membrane.

Ans. The fluid mosaic model, proposed by Singer and Nicolson in 1972, states that the plasma membrane is a fluid phospholipid bilayer with proteins embedded in it in a mosaic pattern. The lipid molecules and many proteins can move laterally, so the membrane is not rigid but flexible and self-sealing. Intrinsic proteins span the bilayer and act as channels, carriers and receptors, while extrinsic proteins lie on the surfaces; cholesterol keeps the membrane stable and glycolipids and glycoproteins act as recognition markers.

Q8. Explain formation, structure and function of lysosomes.

Ans. Lysosomes are formed when vesicles bud off from the Golgi apparatus carrying hydrolytic enzymes made in the RER. Each lysosome is a single-membrane sac containing about forty enzymes that work best in acidic conditions. Their functions are intracellular digestion of food and foreign particles, digestion of worn-out

organelles (autophagy), destruction of excess secretory granules (crinophagy), and defence against invading microbes.

Q9. How does protein signaling differ from steroid signaling?

Ans. Protein (peptide) hormones are water-soluble, so they cannot cross the lipid membrane; they bind receptors on the cell surface and act quickly through a second messenger, cAMP, which activates enzymes inside the cell. Steroid hormones are lipid-soluble, so they pass through the membrane, bind receptors in the cytoplasm or nucleus, and the hormone-receptor complex binds DNA and changes gene expression, producing new proteins more slowly but for a longer time.

Chapter 2: Stem Cell and Cell Cycle

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. Which cell has the ability to divide and differentiate into various cell types?

- ✓ Stem Cell
- Somatic Cell
- Muscle Cell
- Gland Cell

Q2. The stem cell which gives rise to all parts of the body.

- Multipotent
- Pleuripotent
- ✓ Totipotent
- Unipotent

Q3. Which type of stem cell is considered ethically controversial?

- Induced pluripotent stem cells
- ✓ Embryonic stem cells
- Umbilical cord stem cells
- Adult stem cells

Q4. Induced pluripotent stem cells are generated by:

- ✓ Directly reprogramming adult cells
- Extracting stem cells from embryo
- Obtaining stem cells from the bone marrow

Fusing two different types of stem cells

Q5. Which type of stem cell is the most limited kind of cell?

Pluripotent

Multipotent

✓ Unitpotent

Totipotent

Q6. Crossing over occurs during which stage of meiosis?

✓ Prophase-I

Metaphase-I

Anaphase-I

Telophase-I

Q7. If a human skin cell undergoes mitosis, how many chromosomes will each daughter cell contain?

12

23

✓ 46

92

Q8. What is the main difference between cytokinesis in plant and animal cells?

Animal cells form a cell plate

✓ Plant cells form a cell plate (phragmoplast)

Cytokinesis happens only in animals

Cleavage furrow occurs in plant cell

Q9. The structure which divides the plant cell.

- Spindle
- Aster
- ✓ Phragmoplast
- Centrioles

Q10. Why is meiosis important for sexual reproduction?

- It prevents DNA replication
- Increases the chromosomal number
- ✓ Reduces the chromosome number by half
- Doubles the chromosomal number

Exercise - Section II: Short Response Questions

Q1. How is mitosis different from meiosis?

Ans. Mitosis is the division of a body cell that produces two genetically identical diploid daughter cells and occurs for growth and repair. Meiosis is the division of a germ cell that produces four genetically different haploid daughter cells, involves two divisions and crossing over, and occurs for sexual reproduction.

Q2. Differentiate between animal and plant cell mitosis.

Ans. In animal-cell mitosis the spindle has centrioles and asters, and cytokinesis occurs by a cleavage furrow that pinches the cell in two. In plant-cell mitosis there are no centrioles or asters, and cytokinesis occurs by a phragmoplast that builds a cell plate across the middle, which becomes the new cell wall.

Q3. Write the significance of mitosis.

Ans. Mitosis is significant because it produces growth by increasing cell number, replaces worn-out and damaged cells, repairs wounds, maintains the chromosome number in body cells, and allows asexual reproduction in many organisms.

Q4. How is karyokinesis different from cytokinesis?

Ans. Karyokinesis is the division of the nucleus (the separation of chromosomes into two daughter nuclei), while cytokinesis is the division of the cytoplasm that follows it and produces two separate daughter cells. Karyokinesis can occur without cytokinesis, giving a multinucleate cell, but normally the two follow each other.

Q5. What are the preparatory phases of the cell cycle?

Ans. The preparatory phases are the phases of interphase before division: G1 (first gap), in which the cell grows and makes proteins and organelles; S (synthesis), in which DNA is replicated; and G2 (second gap), in which the cell grows further and prepares the machinery for mitosis.

Q6. Briefly state the subphases of prophase-I of meiosis-I.

Ans. Prophase-I of meiosis-I has five subphases: leptotene, when chromosomes become visible as thin threads; zygotene, when homologous chromosomes pair (synapsis); pachytene, when crossing over occurs between non-sister chromatids; diplotene, when homologues begin to separate but stay joined at chiasmata; and diakinesis, when chromosomes condense fully and the nuclear membrane breaks down.

Q7. Define stem cell and their importance.

Ans. A stem cell is an unspecialised cell that can divide to renew itself and can differentiate into specialised cell types. Stem cells are important for growth and development, for replacing damaged and dead cells, and for medical treatments such as bone-marrow transplants and the repair of damaged tissues.

Q8. Differentiate between totipotent and pluripotent stem cells.

Ans. A totipotent stem cell (for example the zygote and the first few embryonic cells) can give rise to a complete organism, including the placenta. A pluripotent stem cell (for example an embryonic stem cell) can give rise to almost all body cell types, but not to a complete organism on its own.

Q9. How are mitosis and meiosis similar?

Ans. Both mitosis and meiosis begin with replication of DNA in interphase, and both pass through the same named stages — prophase, metaphase, anaphase and telophase — with spindle formation and separation of nuclear material. Both also end with cytokinesis, dividing the cytoplasm into daughter cells.

Q10. Why is meiosis-I called the reduction phase?

Ans. Meiosis-I is called the reduction phase because homologous chromosomes separate during anaphase-I, so the chromosome number is halved: one diploid cell produces two haploid cells. Meiosis-II only separates sister chromatids, like mitosis, and does not reduce the number further.

Q11. How can stem cells be harmful?

Ans. Stem cells can be harmful if their division is uncontrolled: they can form tumours (teratomas) when transplanted, and they may differentiate into the wrong tissue or trigger an immune reaction. Embryonic stem cells also raise ethical concerns because obtaining them destroys a human embryo.

Q12. How does G1 differ from G2?

Ans. G1 is the first gap phase after division, when the cell grows, makes RNA, proteins and organelles, and decides whether to divide again. G2 is the second gap phase after DNA replication, when the cell grows a little more, makes the proteins and organelles needed for division, and checks the copied DNA for errors before mitosis begins.

Q13. Why is telophase known as the reversal of prophase?

Ans. Telophase is called the reversal of prophase because everything that happened in prophase is undone: chromosomes uncoil and become thin chromatin again, the spindle disappears, the nuclear membrane and nucleolus reappear, and the cell returns to its interphase appearance.

Q14. Differentiate between metaphase and anaphase.

Ans. In metaphase, chromosomes line up on the equator of the spindle, each attached to spindle fibres at its centromere. In anaphase, the centromeres divide and sister chromatids (or homologous chromosomes in meiosis-I) are pulled apart to opposite poles of the cell.

Q15. Write two advantages of stem cells.

Ans. Two advantages of stem cells are: (1) they can replace damaged or diseased cells and repair tissues, for example in leukaemia treatment by bone-marrow transplant; (2) they can be grown in the laboratory to study diseases and test new drugs without risking patients.

Exercise - Section III: Extended Response Questions

Q1. Define mitosis and explain different phases of mitosis.

Ans. Mitosis is the division of a nucleus that produces two genetically identical daughter cells. Its phases are: Prophase — chromosomes condense, the nuclear membrane and nucleolus disappear and the spindle forms. Metaphase — chromosomes line up on the equator, attached to spindle fibres at their centromeres. Anaphase — centromeres divide and sister chromatids are pulled to opposite poles. Telophase — chromosomes uncoil, the nuclear membrane and nucleolus reappear and the spindle disappears. Cytokinesis then divides the cytoplasm, by a cleavage furrow in animal cells and by a cell plate in plant cells.

Q2. Define meiosis and explain phases of meiosis-I in detail.

Ans. Meiosis is the division of a germ cell that produces four genetically different haploid cells through two divisions. Meiosis-I: Prophase-I (leptotene, zygotene with synapsis, pachytene with crossing over, diplotene, diakinesis) — homologous chromosomes pair and exchange segments. Metaphase-I — homologous pairs line up on the equator. Anaphase-I — homologous chromosomes

separate to opposite poles, halving the number. Telophase-I — two haploid cells form. Meiosis-II then separates sister chromatids in the same four stages as mitosis, giving four haploid cells.

Q3. Describe different types of stem cells.

Ans. Stem cells are of several types. Embryonic stem cells come from the early embryo and are pluripotent. Adult (somatic) stem cells are found in tissues such as bone marrow and skin and are multipotent or unipotent, replacing the cells of their own tissue. Induced pluripotent stem cells (iPSCs) are adult cells reprogrammed in the laboratory to behave like embryonic stem cells. Umbilical-cord and placental stem cells, collected at birth, are also used because they are easy to obtain and ethically uncontroversial.

Q4. What are the advantages and disadvantages of using iPSCs?

Ans. Advantages of iPSCs: they are made from the patient's own cells, so there is little risk of immune rejection; they avoid the ethical problems of embryonic stem cells; and they can be used to model diseases and test drugs. Disadvantages: reprogramming is slow and inefficient; the cells may carry genetic changes or retain memory of their original tissue; and, like embryonic stem cells, they carry a risk of forming tumours if their growth is not controlled.

Q5. Explain different categories of stem cells.

Ans. Stem cells are categorised by their power of differentiation: totipotent cells (the zygote and earliest cells) can form a complete organism; pluripotent cells (embryonic stem cells, iPSCs) can form

nearly all body cell types; multipotent cells (for example blood stem cells) can form several related cell types; oligopotent cells can form a few types; and unipotent cells (for example muscle stem cells) can form only one cell type. They are also categorised by origin: embryonic, adult, cord-blood and induced pluripotent stem cells.

Thinking Deep (model answers)

Q1. If a person needs liver cells, would you use pluripotent cells, or unipotent?

Ans. For liver cells, a unipotent liver stem cell (or a multipotent/pluripotent cell directed to become liver) would be preferred over a general pluripotent cell, because the more specialised the stem cell, the lower the risk that it forms unwanted tissues. In practice, pluripotent cells are often differentiated into liver cells in the laboratory first, and then transplanted.

Q2. Are induced pluripotent stem cells natural?

Ans. No, induced pluripotent stem cells are not natural: they are made in the laboratory by artificially reprogramming ordinary adult cells (for example skin cells) with special genes so that they behave like embryonic stem cells.

Q3. How does mitosis cell division lead to cancer?

Ans. Cancer begins when the control of mitosis is lost: mutations in genes that control the cell cycle let a cell divide again and again without stopping. The extra cells pile up into a tumour, and if they invade other tissues the cancer spreads.

Q4. Compare and contrast the human somatic cell and human gamete.

Ans. A human somatic cell is diploid (46 chromosomes), produced by mitosis, and its daughter cells are genetically identical to it. A human gamete is haploid (23 chromosomes), produced by meiosis, and it is genetically different from the parent cell because of crossing over and independent assortment; two gametes fuse at fertilisation to restore the diploid number.

Chapter 3: Biological Molecules

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. Which of the following is the simplest form of carbohydrates?

- ✓ Monosaccharide
- Disaccharide
- Polysaccharide
- Starch and glycogen

Q2. All are organic substances except.

- Lipids
- ✓ Mineral salts
- Proteins
- Carbohydrates

Q3. Which one of the following groups contains all polysaccharides?

- Sucrose, glucose and fructose
- Maltose, lactose and fructose
- Glycogen, sucrose and maltose
- ✓ Glycogen, cellulose and starch

Q4. Lactose is composed of:

- ✓ Glucose + Galactose
- Fructose + Galactose
- Glucose + Fructose
- Glucose + Glucose

Q5. An ATP molecule is consisting of:

✓ Mono nucleotide

Nucleoside

Polynucleotide

Vitamin

Q6. Lipids are insoluble in water because lipid molecules are:

Hydrophilic

✓ Hydrophobic

Neutral

Polar

Q7. In double helix of DNA, the two DNA strands are:

Coiled around a common axis

✓ Coiled around each other

Coiled differently

Coiled over protein sheath

Q8. Which of the following base is made of double ring?

Thymine

Uracil

✓ Guanine

Cytosine

Q9. The building blocks of all proteins are:

Glucose

Fatty acids

✓ Amino acids

Nucleotide

Q10. Which one is not a conjugated molecule?

Lipoprotein

Glycoprotein

Glycolipid

✓ Polysaccharide

Q11. DNA molecule does not have one of the following sequences.

CCT

CAT

✓ GUA

GAA

Q12. Polyterpene is:

Lanolin

✓ Rubber

Cutin

Suberin

Q13. Phosphatidyl choline is one of the common:

Acylglycerol

✓ Phospholipid

Terpene

Steroid

Q14. The bond in which sharing of electrons between two atoms is

called:

- Ionic Bond
- ✓ Covalent Bond
- Hydrogen Bond
- Hydrophobic Interaction

Q15. NAD is an example of:

- Mononucleotide
- ✓ Dinucleotides
- Polynucleotides
- Trinucleotides

Q16. The bond between two water molecules is:

- Covalent
- Ionic
- Disulphide
- ✓ Hydrogen

Q17. Silk fibre is an example of:

- ✓ Fibrous protein
- Globular protein
- Primary protein
- Tertiary protein

Q18. Which one is an example of oxygen carrying protein?

- Phosphocreatine
- Prothrombin
- Fibrinogen

✓ Haemoglobin

Q19. L and D sugars are an example of:

✓ Enantiomers

Diastereoisomers

Epimer

Disaccharides

Q20. Cutin is an example of:

Steroid

✓ Wax

Terpene

Cholesterol

Exercise - Section II: Short Response Questions

Q1. What are monosaccharides?

Ans. Monosaccharides are the simplest sugars: single sugar units that cannot be broken down into smaller sugars by hydrolysis. They are the building blocks of all carbohydrates; common examples are glucose, fructose and galactose, all with the formula $C_6H_{12}O_6$.

Q2. Write two main functions of carbohydrates in plants.

Ans. In plants, carbohydrates (1) provide stored energy, mainly as starch in seeds, roots and tubers, and (2) provide structural support, mainly as cellulose in cell walls, which is the most abundant organic substance in the plant kingdom.

Q3. What do you understand by the term glycosidic linkage?

Ans. A glycosidic linkage (bond) is the covalent bond formed between two sugar molecules when a water molecule is removed in a condensation reaction. It joins the carbon of one sugar to an oxygen atom and then to the carbon of the next sugar, and it is the bond that links monosaccharides into disaccharides and polysaccharides.

Q4. What is glycogen? How is it different from starch?

Ans. Glycogen is the storage polysaccharide of animals, stored mainly in liver and muscle; it is a highly branched polymer of glucose. Starch is the storage polysaccharide of plants, stored in seeds and tubers; it is a mixture of unbranched amylose (15-20%) and branched amylopectin (80-85%), and it is less branched than glycogen.

Q5. What are the hydrolysis products of sucrose and lactose?

Ans. Hydrolysis of sucrose gives one molecule of glucose and one of fructose. Hydrolysis of lactose gives one molecule of glucose and one of galactose. In both cases a water molecule is added across the glycosidic bond to split the disaccharide.

Q6. What is the basic structural difference between starch and cellulose?

Ans. The basic structural difference is the linkage between glucose units: starch has alpha-1,4 linkages (with alpha-1,6 branches) so its chains coil into helices, while cellulose has beta-1,4 linkages so its chains stay straight and pack into strong, rigid microfibrils. Animals can digest starch but not cellulose.

Q7. How does globular protein differ from fibrous protein?

Ans. Globular proteins fold into compact, roughly spherical shapes, are usually soluble in water, and carry out dynamic jobs such as enzymes, hormones and transport (for example haemoglobin). Fibrous proteins form long, tough fibres, are usually insoluble in water, and serve structural roles (for example collagen, keratin and silk fibroin).

Q8. Define nucleic acids. Mention their two important functions.

Ans. Nucleic acids are long polymers of nucleotides that store and transmit hereditary information. Their two important functions are: (1) DNA stores the genetic instructions of the cell and passes them to the next generation; (2) RNA transfers that information and helps build proteins (mRNA carries the message, tRNA brings amino acids, rRNA forms ribosomes).

Q9. How does nucleoside differ from nucleotide?

Ans. A nucleoside is made of a nitrogenous base joined to a pentose sugar only. A nucleotide is a nucleoside plus one or more phosphate groups; it is the complete building block of nucleic acids, and it can also carry energy (as in ATP) or act as a coenzyme (as in NAD).

Q10. The two strands in DNA are not identical but are complementary. Give reason.

Ans. The two strands are complementary, not identical, because the bases pair by a fixed rule: adenine always pairs with thymine

(two hydrogen bonds) and guanine always pairs with cytosine (three hydrogen bonds). So the sequence of one strand automatically determines the sequence of the other, and the two strands run in opposite (antiparallel) directions.

Q11. Write the important structural and functional differences between DNA and RNA.

Ans. Structurally, DNA is double-stranded, has deoxyribose sugar and the base thymine, and is usually a long helix in the nucleus; RNA is single-stranded, has ribose sugar and the base uracil instead of thymine, and is usually shorter. Functionally, DNA stores genetic information permanently, while RNA copies that information and takes part in protein synthesis.

Q12. Write types of RNA found in the cell.

Ans. The three types of RNA found in the cell are: messenger RNA (mRNA), which carries the genetic message from DNA to the ribosome; transfer RNA (tRNA), which brings the correct amino acid to the ribosome; and ribosomal RNA (rRNA), which forms the structure of ribosomes together with protein.

Q13. How is hydrophobic interaction different from hydrophilic?

Ans. Hydrophobic interaction is the tendency of non-polar molecules (such as the tails of lipids) to stay together and avoid water; it is not a true bond but a result of water pushing non-polar groups together. Hydrophilic means water-loving: polar or charged groups attract water molecules and dissolve or mix with water easily.

Q14. How would you compare mRNA and tRNA?

Ans. mRNA is a long, single-stranded copy of one gene that carries codons from the nucleus to the ribosome and is used up after translation. tRNA is a small, clover-leaf-shaped RNA that carries one specific amino acid on one end and recognises the matching codon on mRNA with its anticodon at the other end.

Q15. Differentiate between pyrimidine and purine.

Ans. Pyrimidines (cytosine, thymine and uracil) are nitrogenous bases made of a single six-membered ring. Purines (adenine and guanine) are made of a double ring, a six-membered ring fused to a five-membered ring. In DNA, a purine always pairs with a pyrimidine, which keeps the width of the double helix constant.

Q16. How does mRNA differ from rRNA?

Ans. mRNA carries the genetic code from DNA to the ribosome as a linear message and is made fresh for each protein. rRNA is a structural and catalytic part of the ribosome itself: together with proteins it forms the ribosome, and it catalyses the formation of peptide bonds during protein synthesis.

Q17. How is hydrogen bond important for existence of life?

Ans. Hydrogen bonds are important for life because they are weak enough to break easily yet strong enough to hold structures together. They hold the two strands of DNA together while still allowing them to separate for copying; they give water its cohesion, high boiling point and solvent power; and they stabilise the shape

of proteins and the folding of RNA.

Q18. What is meant by polarity of water?

Ans. Polarity of water means that its molecule has an uneven distribution of charge: the oxygen atom carries a partial negative charge and the hydrogen atoms carry partial positive charges, because oxygen attracts the shared electrons more strongly. This makes water a polar molecule that dissolves polar and ionic substances and forms hydrogen bonds.

Q19. How would you compare condensation and hydrolysis?

Ans. Condensation is a reaction in which two molecules join together with the removal of a water molecule, for example when two amino acids join by a peptide bond. Hydrolysis is the reverse reaction: a water molecule is added to split a larger molecule into smaller ones, for example when a disaccharide is split into two monosaccharides.

Q20. What do you know about the central dogma of life?

Ans. The central dogma of life states the flow of genetic information: DNA is copied into RNA by transcription, and RNA is used to build proteins by translation — DNA -> RNA -> protein. In some viruses, RNA can also be copied back into DNA by reverse transcription.

Q21. Define peptide bond.

Ans. A peptide bond is the covalent bond that joins two amino acids: it forms between the carboxyl group (-COOH) of one amino

acid and the amino group (-NH2) of the next, with the removal of a water molecule. A chain of amino acids joined by peptide bonds is called a polypeptide.

Q22. What is the significance of sequence of amino acids in a protein molecule?

Ans. The sequence of amino acids (the primary structure) determines how the protein folds into its secondary, tertiary and quaternary shapes, and its shape determines its function. Even one changed amino acid can alter the shape and destroy the function, as in sickle-cell haemoglobin, where one substitution changes the red blood cell's shape.

Q23. What are conjugated molecules? Give examples.

Ans. Conjugated molecules are biological molecules formed when a simple molecule (such as a protein, lipid or carbohydrate) is combined with a non-carbohydrate or non-protein group. Examples are glycoproteins (protein + carbohydrate), glycolipids (lipid + carbohydrate), lipoproteins (lipid + protein) and nucleoproteins (protein + nucleic acid).

Exercise - Section III: Extended Response Questions

Q1. How are amino acids linked to form proteins?

Ans. Amino acids are linked by peptide bonds: the carboxyl group of one amino acid reacts with the amino group of the next, releasing a water molecule (a condensation reaction). Repeated many times this builds a long polypeptide chain. The chain then folds into secondary structures (alpha-helix and beta-sheet), then into a

three-dimensional tertiary shape, and sometimes several chains join into a quaternary structure; the final shape is the working protein.

Q2. What are the three components of a nucleotide? How are nucleotides linked to form nucleic acids?

Ans. A nucleotide has three components: a nitrogenous base (adenine, guanine, cytosine, thymine or uracil), a pentose sugar (deoxyribose in DNA, ribose in RNA) and one or more phosphate groups. Nucleotides are linked by phosphodiester bonds: the phosphate of one nucleotide joins the 3' carbon of one sugar to the 5' carbon of the next sugar, forming a sugar-phosphate backbone with the bases pointing inward, as in the DNA double helix.

Q3. What are the two kinds of subunits that make up a neutral fat molecule, and how are they arranged in the molecule?

Ans. A neutral fat (triglyceride) molecule is made of two kinds of subunits: one glycerol molecule and three fatty acids. The glycerol forms the backbone, and each of its three hydroxyl groups is joined to the carboxyl group of a fatty acid by an ester bond. The fatty acids may be saturated (no double bonds) or unsaturated (one or more double bonds), which affects whether the fat is solid or liquid.

Q4. What is a polysaccharide? Explain its types.

Ans. A polysaccharide is a large carbohydrate made of many monosaccharide units joined by glycosidic bonds. Its types are: storage polysaccharides — starch in plants (amylose + amylopectin) and glycogen in animals (highly branched); and

structural polysaccharides — cellulose in plant cell walls (beta-1,4 linked glucose) and chitin in fungal walls and arthropod exoskeletons.

Q5. Describe structure and function of protein.

Ans. A protein's structure has four levels: primary — the sequence of amino acids; secondary — local folding into alpha-helices and beta-sheets held by hydrogen bonds; tertiary — the full three-dimensional folding of one chain; and quaternary — the joining of several chains. Its functions include enzymes that speed up reactions, structural support (collagen, keratin), transport (haemoglobin), defence (antibodies), movement (actin, myosin) and regulation (hormones such as insulin).

Q6. Analyze the characters of steroids and prostaglandins.

Ans. Steroids are lipids built from four fused carbon rings; they include cholesterol (which stabilises membranes and is the starting material for other steroids), sex hormones (testosterone, oestrogen) and vitamin D. Prostaglandins are lipid compounds made from fatty acids; they act locally as chemical messengers, causing inflammation, pain and fever, controlling blood clotting and regulating smooth-muscle contraction. Both are non-polar and insoluble in water, but steroids act mainly through gene regulation while prostaglandins act through surface receptors.

Q7. Explain the Watson and Crick model of DNA.

Ans. The Watson-Crick model (1953) states that DNA is a double helix of two antiparallel strands coiled around a common axis. Each

strand has a sugar-phosphate backbone on the outside and nitrogenous bases on the inside; adenine pairs with thymine (two hydrogen bonds) and guanine pairs with cytosine (three hydrogen bonds). The model explains how DNA is copied (each strand acts as a template) and how genetic information is stored in the sequence of bases.

Activity (model answers)

Q1. Label the diagram of macromolecules and write their sub units.

Ans. The four macromolecules and their subunits are: lipids — fatty acids and glycerol (shown as membrane bilayers and droplets); nucleic acids — nucleotides (shown as the DNA double helix); polysaccharides — monosaccharides such as glucose (shown as the branched starch/glycogen molecule); and proteins — amino acids (shown as the folded polypeptide chain).

Q2. Draw the Watson-Crick and Wilkins model of DNA.

Ans. The Watson-Crick-Wilkins model of DNA is a double helix: two antiparallel sugar-phosphate backbones on the outside, with A-T and G-C base pairs joined by hydrogen bonds on the inside, one full turn every 10 base pairs. Draw the ladder twisted into a helix and label: sugar-phosphate backbone, adenine, thymine, guanine, cytosine and hydrogen bonds.

Q3. Make a chart showing comparison between DNA and RNA. Hypothesize which organic substance came first.

Ans. DNA and RNA comparison: DNA is double-stranded, has deoxyribose and thymine, and stores information; RNA is

single-stranded, has ribose and uracil, and works in protein synthesis (mRNA carries the message, tRNA delivers amino acids, rRNA builds ribosomes). As a hypothesis, the first organic substances were probably simple molecules such as amino acids and nucleotides, formed from inorganic gases and later joined into polymers — the heterotrophic origin-of-life idea.

Chapter 4: Enzymes

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. Which feature allows enzymes to be reusable in living systems?

- ✓ They are not consumed during reactions
- They can synthesize new substrates
- They act as both reactants and products
- They lower activation energy permanently

Q2. An enzyme with its coenzyme or prosthetic group is called as:

- ✓ Holoenzyme
- Apoenzyme
- Activator
- Inhibitor

Q3. Generally a single enzyme catalyzes only a single substrate or a group of related substrates, therefore, the enzymes are:

- ✓ Specific
- Reactive
- Activator
- Inhibitor

Q4. The enzymes involved in cellular respiration are found in:

- Golgi bodies
- ✓ Mitochondria
- Chloroplast

Ribosomes

Q5. Every enzyme functions most effectively over a narrow range of pH known as:

Maximum

Minimum

✓ Optimum

Both a and b

Q6. At very high temperatures, enzymes lose activity because:

Substrate concentration decreases

✓ They are denatured and lose their 3D shape

Activation energy increases

Molecular movement decreases

Q7. The chemical substance with which an enzyme reacts is called its:

✓ Substrate

Active site

Inhibitor

Cofactor

Q8. Which of the following best explains the induced fit model of enzyme action?

Enzyme's active site is rigid and unchangeable

✓ Substrate causes a conformational change in enzyme, improving binding

Substrate breaks apart before binding to enzyme

Enzyme is consumed during the reaction

Q9. The optimum temperature for enzymes in human body is:

36 degrees C

✓ 37 degrees C

38 degrees C

39 degrees C

Q10. The catalytic activity of an enzyme is restricted to its small portion called:

✓ Active site

Passive site

Binding site

Allosteric site

Q11. The reversible inhibitors usually constitute:

Strong linkage with enzyme

✓ Weak linkage with enzyme

No linkage with enzyme

Medium linkage with enzyme

Q12. A non-competitive inhibitor differs from a competitive one because it:

Competes with substrate for the active site

✓ Lowers V-max

Increases V-max

Prevents substrate binding by occupying the active site

Q13. Enzymes lower the:

✓ Activation Energy

Potential Energy

Kinetic Energy

Ionic Energy

Q14. The term apoenzyme is applicable to:

Single Enzyme

✓ Protein Part of Enzyme

Organic Co-factor

Inorganic Co-factor

Q15. The enzymes that breakdown polymer into monomers by adding water are called:

Transferases

Lyases

✓ Hydrolases

Ligases

Exercise - Section II: Short Response Questions

Q1. How does temperature affect enzyme activity?

Ans. At low temperatures enzyme activity is slow because molecules move slowly and collide less often. As temperature rises, activity increases up to the optimum temperature (about 37 degrees C in humans). Beyond the optimum, the enzyme's shape is destroyed (denaturation) and activity falls sharply to zero.

Q2. How does pH affect enzyme activity?

Ans. Every enzyme works best at a particular pH, its optimum pH; for example pepsin works best at pH 2 in the stomach and trypsin at pH 8 in the intestine. If the pH moves away from the optimum, the charges on the enzyme change, its shape is disturbed, and activity falls; at extreme pH the enzyme is denatured.

Q3. What is meant by energy of activation?

Ans. Energy of activation is the minimum amount of free energy that reactant molecules must gain before a chemical reaction can start. It is the energy barrier that must be crossed for bonds to break and new bonds to form; enzymes work by lowering this barrier.

Q4. How is feedback inhibition important?

Ans. Feedback inhibition is important because it prevents waste: when the end product of a metabolic pathway accumulates, it binds to the first enzyme of the pathway and switches it off. This stops the cell from making more product than it needs and keeps the internal environment stable.

Q5. What is meant by optimum temperature and optimum pH?

Ans. Optimum temperature is the temperature at which an enzyme shows its maximum activity (about 37 degrees C for human enzymes). Optimum pH is the pH at which an enzyme shows its maximum activity (for example pH 2 for pepsin, pH 7.5 for most intracellular enzymes). On either side of the optimum, activity decreases.

Q6. Write the names of six classes of enzymes on the basis of reaction they perform.

Ans. The six classes of enzymes on the basis of the reaction they perform are: (1) Oxidoreductases — oxidation-reduction reactions; (2) Transferases — transfer of a group from one molecule to another; (3) Hydrolases — hydrolysis (breaking bonds by adding water); (4) Lyases — breaking bonds without water or oxidation; (5) Isomerases — rearrangement within a molecule; (6) Ligases — joining two molecules using ATP.

Q7. What is the role of active site of enzyme?

Ans. The active site is the small pocket on the enzyme where the substrate binds and the reaction takes place. Its shape fits the substrate, and it contains the binding site that holds the substrate and the catalytic site where bonds are broken or formed.

Q8. How are three types of co-factors different?

Ans. The three types of cofactors differ as follows: inorganic cofactors (metal ions such as Zn2+, Mg2+) are loosely attached and activate the enzyme; coenzymes are small organic molecules (often vitamins, such as NAD and FAD) that carry groups between reactions; prosthetic groups are organic cofactors tightly and permanently bound to the enzyme (such as haem in catalase).

Q9. How does induced fit model differ from lock and key model of enzyme action?

Ans. In the lock-and-key model (Fischer, 1894), the enzyme's active site is rigid and the substrate fits into it exactly like a key into a

lock, with no change in shape. In the induced-fit model (Koshland, 1959), the active site is flexible: it changes shape slightly when the substrate binds, so the fit becomes tighter and catalysis is improved.

Q10. What are the regulatory and non-regulatory enzymes?
Ans. Regulatory enzymes are enzymes whose activity is controlled, for example allosteric enzymes that are switched on or off by effector molecules, and enzymes controlled by feedback inhibition. Non-regulatory enzymes are ordinary enzymes that simply catalyse their reaction whenever substrate is present, without such control.

Q11. Find out differences between hydrolases and ligases.
Ans. Hydrolases break a bond by adding a water molecule, for example proteases breaking proteins into amino acids. Ligases do the opposite: they join two molecules together, using energy from ATP, for example DNA ligase joining DNA fragments. So hydrolases split molecules, while ligases build them.

Q12. Differentiate the following: (i) Hydrolases and Lyases (ii) Reversible and irreversible inhibitor (iii) Apoenzyme and coenzyme (iv) Active site and allosteric site (v) Competitive and non-competitive enzyme inhibition.
Ans. (i) Hydrolases split bonds by adding water, while lyases split bonds without adding water or oxidation, often forming a double bond. (ii) A reversible inhibitor binds weakly and temporarily, so the enzyme recovers when the inhibitor leaves; an irreversible inhibitor binds strongly or covalently and permanently destroys the active

site. (iii) An apoenzyme is the protein part of the enzyme alone (inactive), while a coenzyme is the small organic non-protein part that completes it. (iv) The active site is where the substrate binds and reacts; the allosteric site is a separate site where a regulator binds and changes the enzyme's shape and activity. (v) A competitive inhibitor competes with the substrate for the active site, while a non-competitive inhibitor binds elsewhere, changes the enzyme's shape and lowers V-max.

Q13. What are thermolabile biocatalysts?

Ans. Thermolabile biocatalysts are enzymes that are destroyed by heat: because they are proteins, high temperature breaks the weak bonds holding their three-dimensional shape, so they lose their active site and their catalytic activity. Most human enzymes are thermolabile and are denatured above about 45 degrees C.

Q14. What is papain enzyme?

Ans. Papain is a proteolytic (protein-digesting) enzyme obtained from the unripe fruit of the papaya plant. It breaks down proteins into peptides and amino acids, and it is used to tenderise meat, in food processing, and in medicines for digestion.

Q15. How do lyases differ from hydrolases?

Ans. Lyases break chemical bonds without hydrolysis and without oxidation, often by removing a group and forming a double bond (for example decarboxylases). Hydrolases break bonds by adding a molecule of water across the bond (for example proteases and lipases). So the difference is whether water is used in the splitting.

Q16. What is the chemical nature of ribozyme?

Ans. A ribozyme is chemically an RNA molecule, not a protein. It is a catalytic RNA that can cut or join other RNA molecules, showing that biological catalysis is not limited to protein enzymes.

Q17. What is catalysis of enzyme?

Ans. Catalysis of enzyme means the speeding up of a chemical reaction by an enzyme without the enzyme itself being used up. The enzyme lowers the activation energy by binding the substrate at its active site, forming an enzyme-substrate complex, converting it to products, and then releasing the unchanged enzyme to work again.

Q18. Define maltase and sucrase.

Ans. Maltase is the enzyme that hydrolyses maltose into two molecules of glucose. Sucrase (invertase) is the enzyme that hydrolyses sucrose into glucose and fructose. Both are carbohydrases found in the small intestine, and their names follow the rule of adding '-ase' to the substrate.

Exercise - Section III: Extended Response Questions

Q1. Describe the physical and chemical structure of an enzyme.

Ans. Physically, an enzyme is a globular protein with a specific three-dimensional shape, folded into a tertiary (sometimes quaternary) structure; its shape creates the active site. Chemically, most enzymes are proteins, but many need a non-protein cofactor to work: the protein part alone is the apoenzyme, and apoenzyme

+ cofactor = holoenzyme. Cofactors may be metal ions, coenzymes (organic, often vitamins) or prosthetic groups tightly bound to the protein. Some catalytic RNAs (ribozymes) are exceptions to the protein rule.

Q2. Describe the two models about the mechanism of enzyme action.

Ans. The lock-and-key model (Emil Fischer, 1894) says the active site is rigid and the substrate fits it exactly, like a key in a lock, so only one substrate fits each enzyme. The induced-fit model (Daniel Koshland, 1959) says the active site is flexible: when the substrate approaches, the enzyme changes shape slightly so that the binding and catalytic groups are brought into the right position. The induced-fit model is now accepted because it explains how enzymes can act on slightly different substrates and how binding can strain the substrate's bonds.

Q3. Explain different factors which affect enzyme activity.

Ans. Enzyme activity is affected by: temperature — activity rises to an optimum (about 37 degrees C in humans) and then falls as the enzyme denatures; pH — each enzyme has an optimum pH and is denatured at extremes; substrate concentration — activity rises with substrate until all active sites are saturated (V-max); enzyme concentration — more enzyme gives faster reaction if substrate is abundant; inhibitors — competitive inhibitors block the active site, non-competitive inhibitors bind elsewhere; and cofactors/activators — many enzymes need metal ions or coenzymes to work.

Q4. Define enzyme inhibition and describe different types of enzyme inhibitors in detail.

Ans. Enzyme inhibition is the reduction or stopping of enzyme activity by a substance called an inhibitor. Reversible inhibition: the inhibitor binds weakly and temporarily — competitive inhibitors resemble the substrate and compete for the active site (raising Km but not changing V-max), while non-competitive inhibitors bind at another site and lower V-max. Irreversible inhibition: the inhibitor binds strongly or covalently to the active site and permanently destroys the enzyme, for example nerve gases and some antibiotics. Feedback inhibition is a special reversible control in which the end product of a pathway inhibits its first enzyme.

Q5. Write a detailed note on classification of enzymes on the basis of their reaction type.

Ans. On the basis of reaction type, enzymes are classified into six classes: (1) Oxidoreductases catalyse oxidation-reduction (for example dehydrogenases); (2) Transferases transfer a chemical group (for example kinases); (3) Hydrolases split bonds by adding water (proteases, lipases, carbohydrases); (4) Lyases split bonds without water, forming double bonds (decarboxylases); (5) Isomerases rearrange atoms within a molecule (for example phosphohexose isomerase); (6) Ligases join molecules using ATP (for example DNA ligase). Enzymes are also named by substrate (protease, lipase), by reaction (oxidase, reductase), or by both (DNA polymerase).

Q6. Write the catalytic characteristics of enzymes.

Ans. The catalytic characteristics of enzymes are: they are specific, acting on one substrate or a group of related substrates; they are efficient, speeding up reactions millions of times; they are reusable, emerging unchanged after each reaction; they work in small amounts; they lower the activation energy; they are sensitive to temperature and pH and are denatured by extremes; and their activity can be regulated by activators, inhibitors and feedback control.

Q7. Describe effect of temperature and pH on enzyme action.

Ans. Effect of temperature: as temperature rises, molecular movement and collisions increase, so the reaction rate rises up to the optimum (about 37 degrees C in humans); above the optimum the protein denatures and activity drops sharply. Effect of pH: each enzyme has an optimum pH at which its shape and charges are correct; moving away from it reduces binding, and extreme pH denatures the enzyme. For example, pepsin works in the stomach at pH 2, while trypsin works in the intestine at pH 8.

Q8. Define and explain energy of activation and feedback inhibition.

Ans. Energy of activation is the minimum energy reactants must gain to start a reaction; enzymes lower it by bringing substrates together in the right orientation and straining their bonds at the active site, so more molecules can react at body temperature. Feedback inhibition is the control of a metabolic pathway by its own end product: when the product is abundant it binds to the first enzyme of the pathway (usually at an allosteric site) and switches it

off, preventing overproduction; when the product is used up, the enzyme works again.

Q9. What are cofactors? Discuss their different types.

Ans. Cofactors are non-protein substances that many enzymes need in order to work. Their types are: (1) Inorganic cofactors (metal ions such as Zn2+, Mg2+, Fe2+) loosely attached to the enzyme; (2) Coenzymes — small organic molecules, often derived from vitamins, that are loosely bound and carry chemical groups (for example NAD, FAD, coenzyme A); (3) Prosthetic groups — organic cofactors tightly and permanently bound to the enzyme (for example haem in haemoglobin and catalase, and FAD in some enzymes). The complete active enzyme (apoenzyme + cofactor) is called the holoenzyme.

Q10. Compare between reversible and irreversible enzyme inhibition.

Ans. Reversible inhibition: the inhibitor binds weakly (by hydrogen bonds or ionic bonds) and temporarily, so the enzyme recovers when the inhibitor is removed; it includes competitive inhibition (at the active site) and non-competitive inhibition (at another site). Irreversible inhibition: the inhibitor binds strongly or covalently and permanently alters or destroys the active site, so the enzyme cannot recover; examples are cyanide, nerve gases and penicillin (which blocks a bacterial enzyme permanently).

Activity (model answer)

Q1. Label the alphabets (a to g) in the enzyme action diagram with

suitable words.

Ans. In the enzyme-action diagram: (a) is the substrate (the two joined sugar units); (b) and (c) are the enzyme (the purple shape); (d) are the products after the substrate has been split; (e) marks the active site region where the substrate binds; (f) is the enzyme-substrate complex; and (g) marks the catalytic site within the active site where the bond is broken. (Labels read from the diagram; the two white spots mark the active-site region of the enzyme.)

Thinking Deep (model answers)

Q1. How do enzymes increase the rate of chemical reaction?

Ans. Enzymes increase the rate of a reaction by lowering its activation energy: they bind the substrate at the active site, hold the reacting molecules in the correct orientation, and strain the bonds to be broken, so many more molecules have enough energy to react at body temperature.

Q2. Explain how exercise involves both anabolic and catabolic reactions.

Ans. Exercise involves anabolic reactions because muscles build up glycogen stores and repair and build muscle protein after work. It involves catabolic reactions because muscles break down glucose and glycogen by respiration to release the ATP that powers contraction.

Q3. How does enzyme feedback inhibition benefit a cell?

Ans. Feedback inhibition benefits the cell by preventing waste and

keeping balance: it stops a pathway as soon as enough end product has been made, so raw materials and energy are not wasted, and it keeps the concentrations of metabolites steady (homeostasis).

Q4. With regard to enzymes why are vitamins and minerals necessary?

Ans. Vitamins and minerals are necessary because many of them are cofactors: B-vitamins form coenzymes such as NAD and FAD, and minerals such as zinc, magnesium and iron act as inorganic cofactors or parts of prosthetic groups. Without them the enzymes cannot work properly, and deficiency diseases appear.

Q5. What are the differences between lock and key mechanism and induced fit mechanism of enzyme action?

Ans. In the lock-and-key model the active site is rigid and the substrate fits exactly without any change; in the induced-fit model the active site is flexible and changes shape when the substrate binds, giving a tighter fit and better catalysis.

Chapter 5: Bioenergetics

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. In photosystem-I the first electron acceptor is:

- Plastocyanin
- Cytochrome complex
- ✓ Ferredoxin
- Iron-sulphur protein

Q2. During the dark reaction of photosynthesis:

- Water is split off
- ✓ CO_2 is reduced to organic compounds
- Chlorophyll is activated
- Glucose is broken down

Q3. The primary enzyme that fixes atmospheric CO_2 in C4 plants is:

- ✓ PEP carboxylase
- Rubisco
- RuBP carboxylase
- Hydrogenase

Q4. The number of carbon atoms in RuBP which accepts CO_2 in C3 plants is:

- 2
- 3
- ✓ 5

6

Q5. Chlorophyll a differs from chlorophyll b in having a:

-CHO group

-COOH group

✓ -CH3 group

-NH2 group

Q6. NADP is:

An enzyme

A part of rRNA

✓ A coenzyme

A part of tRNA

Q7. The compound that enters the Krebs cycle from glycolysis is:

Citric acid

Oxaloacetate

Pyruvic acid

✓ Acetyl coenzyme A

Q8. Carotenoids are:

Proteins

✓ Terpenoid Lipid

Necessary pigment

Protein and terpenoid

Q9. During process of photosynthesis water acts as:

Oxidizing agent

- ✓ Reducing agent
- Both oxidizing and reducing agent
- None of these

Q10. In the leaves of C4 plants malate formation occurs in the cells of:

- Epidermis
- Phloem
- ✓ Mesophyll
- Bundle sheath cells

Q11. ATP synthesis during light dependent reaction is:

- Oxidative phosphorylation
- ✓ Photophosphorylation
- Photolysis
- Substrate level of phosphorylation

Q12. Chlorophylls are soluble in:

- ✓ Organic solvent
- Water
- Water and Organic solvent
- All of these

Q13. The CO2 fixation in C4 plant occurs in:

- Phloem tissue
- Palisade tissue
- Spongy cells
- ✓ Spongy mesophyll and bundle sheath cell

Q14. ATP is consumed during:

✓ Glycolysis

Light dependent phase

Krebs cycle

ETC

Q15. In C3 plants the first stable product of dark reaction of photosynthesis is:

✓ PGA

G3P

RuBP

Oxaloacetate

Exercise - Section II: Short Response Questions

Q1. Write few lines about absorption spectrum.

Ans. The absorption spectrum of a pigment is a graph showing how much light of each wavelength the pigment absorbs. Chlorophyll a absorbs most strongly in the blue (about 430 nm) and red (about 660 nm) regions, and absorbs green light poorly, which is why leaves look green. The action spectrum, by contrast, shows the rate of photosynthesis at each wavelength.

Q2. Write briefly the role of CO_2 in photosynthesis.

Ans. Carbon dioxide is the raw material that is reduced in the dark reaction (Calvin cycle) to form organic compounds. It combines with RuBP to form PGA, which is then reduced using ATP and NADPH to G3P and finally to glucose and other carbohydrates; without CO_2 no

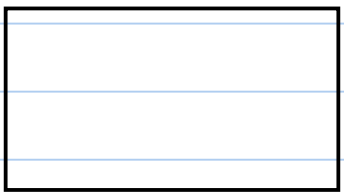
sugar can be made.

Q3. Write the names of electron acceptors of PS II and PS I.

Ans. The electron acceptors are: Photosystem II (P680) — its first acceptor is pheophytin, which passes electrons to plastoquinone; Photosystem I (P700) — its acceptor chain passes electrons through iron-sulphur proteins to ferredoxin, which finally reduces NADP+ to NADPH.

Q4. Draw the Z-scheme of light reaction.

Ans. The Z-scheme is the zig-zag diagram of the light reactions: light excites P680 in PS II, electrons pass downhill through plastoquinone, the cytochrome complex and plastocyanin (releasing energy used to pump protons), then light excites P700 in PS I, and electrons pass through ferredoxin to NADP+, forming NADPH. Water is split at PS II to replace the lost electrons, releasing oxygen. (Draw the zig-zag path: P680 up, downhill to P700, up again, downhill to NADP+.)



Z-scheme: P680 (PS II) up, downhill via plastoquinone and cytochrome to P700 (PS I), up again, downhill via ferredoxin to NADP+ (NADPH); water split at PS II gives O2

Q5. What do you know about the reduction phase of Calvin cycle?

Ans. In the reduction phase of the Calvin cycle, the PGA formed by carbon fixation is reduced to G3P (glyceraldehyde-3-phosphate) using ATP and NADPH from the light reactions. For every three CO2 fixed, six G3P are made; one G3P leaves the cycle to make glucose and other sugars, while the other five are used to regenerate RuBP.

Q6. Define aerobic and anaerobic respiration.

Ans. Aerobic respiration is the breakdown of glucose using oxygen, giving carbon dioxide, water and a large amount of energy (about 36-38 ATP per glucose). Anaerobic respiration (fermentation) is the breakdown of glucose without oxygen, giving much less energy (2 ATP) and products such as lactic acid in muscles or ethanol and carbon dioxide in yeast.

Q7. Write first three steps of Krebs cycle.

Ans. The first three steps of the Krebs (citric acid) cycle are: (1) Acetyl-CoA (2C) joins oxaloacetate (4C) to form citrate (6C). (2) Citrate is rearranged to isocitrate. (3) Isocitrate is oxidised to alpha-ketoglutarate (5C), releasing CO2 and forming NADH.

Q8. What is the importance of PGAL?

Ans. PGAL (phosphoglyceraldehyde, G3P) is important because it is the first carbohydrate product of the Calvin cycle that leaves the cycle. Two PGAL molecules combine to form glucose, and PGAL is also the starting material for starch, sucrose, fats and amino acids, and five of every six PGAL molecules are used to regenerate RuBP.

Q9. How is acetyl CoA formed from the fats?

Ans. Fats are first broken down into glycerol and fatty acids. The fatty acids are then broken down by beta-oxidation: two-carbon pieces are cut off one by one, and each piece joins coenzyme A to form acetyl-CoA, which enters the Krebs cycle to release energy.

Q10. What is the difference between reaction centre and antenna complex?

Ans. The reaction centre is the small group of special chlorophyll molecules (P680 in PS II, P700 in PS I) where light energy is converted into chemical energy by ejecting an excited electron. The antenna complex is the larger group of surrounding pigment molecules (chlorophyll b, carotenoids) that absorb light and funnel the energy to the reaction centre.

Q11. Differentiate between cyclic and non-cyclic photophosphorylation.

Ans. In cyclic photophosphorylation, electrons from PS I return to PS I in a cycle, producing only ATP and no NADPH or oxygen. In non-cyclic photophosphorylation, electrons flow from water through PS II and PS I to NADP+, producing ATP, NADPH and oxygen; this is the main pathway of the light reactions.

Q12. How does regeneration of RuBP occur?

Ans. RuBP is regenerated in the last phase of the Calvin cycle: of the six G3P molecules made from three CO2, five are rearranged through a series of reactions using ATP into three molecules of RuBP (5C). The regenerated RuBP then accepts new CO2 molecules,

so the cycle can continue.

Q13. How does alcoholic fermentation take place?

Ans. In alcoholic fermentation, glucose is first broken down by glycolysis into pyruvate (2 ATP net). In the absence of oxygen, pyruvate is converted to acetaldehyde (releasing CO_2) and then reduced to ethanol, regenerating NAD+ so that glycolysis can continue. Yeast carries out this process, used in brewing and baking.

Q14. Differentiate between chlorophyll a and chlorophyll b.

Ans. Chlorophyll a has a methyl (-CH_3) group at one position and is the main pigment: it is present in all photosynthetic organisms and forms the reaction centres P680 and P700. Chlorophyll b has an aldehyde (-CHO) group at the same position, is found only in the antenna complex, absorbs light of slightly different wavelengths and passes the energy to chlorophyll a.

Q15. Define carotenoids.

Ans. Carotenoids are yellow, orange and red accessory pigments, chemically terpenoid lipids. They absorb blue and green light and pass the energy to chlorophyll, and they protect chlorophyll from damage by excess light by absorbing harmful energy and acting as antioxidants.

Exercise - Section III: Extended Response Questions

Q1. Explain the mechanism of photosynthesis.

Ans. Photosynthesis has two stages. (1) Light reactions (in the

thylakoids): chlorophyll absorbs light; water is split (photolysis), releasing oxygen; electrons flow through PS II and PS I (Z-scheme), and ATP (photophosphorylation) and NADPH are formed. (2) Dark reactions (Calvin cycle, in the stroma): CO2 is fixed by RuBP to form PGA, PGA is reduced by ATP and NADPH to G3P, one G3P in six leaves to make glucose, and the rest regenerate RuBP. Overall: 6CO2 + 12H2O -> C6H12O6 + 6O2 + 6H2O.

Q2. What are the main events of glycolysis? How is ATP formed?

Ans. Glycolysis occurs in the cytoplasm and splits one glucose (6C) into two pyruvate (3C) molecules through ten steps, giving a net gain of 2 ATP and 2 NADH. ATP is formed in two ways: substrate-level phosphorylation, where a phosphate is transferred directly from an intermediate to ADP (in glycolysis and the Krebs cycle); and oxidative phosphorylation, where NADH and FADH2 pass electrons along the electron transport chain, pumping protons that drive ATP synthase (chemiosmosis).

Q3. Describe the role of water in photosynthesis.

Ans. Water plays three roles in photosynthesis: (1) it is the source of electrons that replace those lost by chlorophyll in PS II; (2) its splitting (photolysis) releases oxygen as a by-product; and (3) it provides the protons (H+) that build up inside the thylakoid and drive ATP formation by chemiosmosis. Without water, the light reactions stop and no ATP or NADPH can be made.

Q4. What are the main events of the citric acid cycle?

Ans. The citric acid (Krebs) cycle occurs in the mitochondrial

matrix. Acetyl-CoA (2C) joins oxaloacetate (4C) to form citrate (6C); citrate is oxidised step by step, releasing two CO_2 molecules and regenerating oxaloacetate. Each turn produces 3 NADH, 1 $FADH_2$ and 1 ATP (GTP) by substrate-level phosphorylation. Since one glucose gives two acetyl-CoA, the cycle turns twice per glucose.

Q5. Describe different steps of Electron Transport Chain.

Ans. The electron transport chain is a series of carriers (NADH dehydrogenase, cytochrome complexes, coenzyme Q, cytochrome c, oxygen as final acceptor) in the inner mitochondrial membrane. NADH and $FADH_2$ donate electrons, which pass downhill from carrier to carrier; the released energy pumps protons (H^+) into the intermembrane space. Oxygen accepts the electrons at the end and joins protons to form water. The proton gradient then drives ATP synthase to make ATP (chemiosmosis).

Q6. Explain non-cyclic photophosphorylation.

Ans. Non-cyclic photophosphorylation is the main light-reaction pathway: light excites PS II (P680); electrons from split water pass through plastoquinone, the cytochrome complex and plastocyanin to PS I (P700), which is also excited by light; electrons then pass through ferredoxin to $NADP^+$, forming NADPH. As electrons flow, protons are pumped into the thylakoid lumen, and their flow back through ATP synthase forms ATP. Oxygen is released from water. The products are ATP, NADPH and O_2.

Q7. Describe in detail Dark Reaction.

Ans. The dark reaction (Calvin cycle) occurs in the stroma and

needs no light directly, but uses ATP and NADPH from the light reactions. It has three phases: (1) Carbon fixation — CO2 joins RuBP (5C) by rubisco to form two PGA (3C) molecules. (2) Reduction — PGA is reduced to G3P using ATP and NADPH. (3) Regeneration — five of every six G3P are rearranged, using ATP, into RuBP, while one G3P leaves to form glucose and other sugars. Three turns of the cycle fix three CO2 and produce one G3P net.

Q8. Explain mechanism of Chemiosmosis.

Ans. Chemiosmosis is the process by which ATP is made using a proton gradient. In photosynthesis, the light reactions pump protons (H+) into the thylakoid lumen; in respiration, the electron transport chain pumps protons into the intermembrane space of the mitochondrion. The protons can only flow back through the enzyme ATP synthase, embedded in the membrane; this flow turns the enzyme and provides the energy to join ADP and inorganic phosphate into ATP. This coupling of electron flow to ATP formation was proposed by Mitchell.

Q9. What do you know about photorespiration?

Ans. Photorespiration is a wasteful process that occurs in C3 plants in bright light and high temperature, when rubisco binds oxygen instead of carbon dioxide. Oxygen joins RuBP, forming one PGA and one phosphoglycolate (2C); the phosphoglycolate is processed through peroxisomes and mitochondria, releasing CO2. The plant thus loses fixed carbon and energy without making sugar. C4 plants avoid photorespiration by first fixing CO2 with PEP carboxylase, which has no affinity for oxygen.

Q10. Explain photosynthesis in C4 plant.

Ans. In C4 plants (for example maize, sugarcane), CO2 is first fixed in the mesophyll cells by PEP carboxylase into oxaloacetate (4C), which is converted to malate. Malate moves into the bundle-sheath cells, where it releases CO2 at a high concentration around rubisco, so the Calvin cycle runs efficiently and photorespiration is avoided. The remaining pyruvate returns to the mesophyll cells to regenerate PEP, using ATP. This extra pathway (the Hatch-Slack cycle) lets C4 plants photosynthesise well in hot, dry conditions.

Activity (model answers)

Q1. Compare photosynthesis and cellular respiration by labeling the boxes and completing the sentences.

Ans. Labels: the sun box = Sunlight; the green organelle = Chloroplasts (photosynthesis); the red organelle = Mitochondria (cellular respiration); the inputs to the chloroplast = Water and Carbon Dioxide; the outputs of photosynthesis = Glucose and Oxygen; the inputs to the mitochondrion = Glucose and Oxygen; the outputs of respiration = Carbon Dioxide, Water and ATP (energy). (1) The products of photosynthesis are the inputs of cellular respiration. (2) The products of cellular respiration are the inputs of photosynthesis. (3) The purpose of photosynthesis is to capture light energy and store it as glucose. (4) The purpose of cellular respiration is to release energy (ATP) from glucose. (5) Photosynthesis is done by chloroplasts. (6) Cellular respiration is done by mitochondria.

Thinking Deep (model answers)

Q1. Why can ATP not be stored in excess?

Ans. ATP cannot be stored in excess because it is unstable: it breaks down quickly into ADP, releasing its energy, and the cell would waste energy keeping a large stock. Instead, cells store energy as stable molecules (starch in plants, glycogen and fats in animals) and make ATP on demand when needed.

Q2. Why does caffeine give you so much energy, while being so low in calories?

Ans. Caffeine gives energy without calories because it does not supply fuel; it blocks adenosine receptors in the brain, so the feeling of tiredness is masked, and it stimulates the release of adrenaline, which increases alertness and the breakdown of stored fuel.

Q3. Does the body lose or gain energy when we drink or eat hot things?

Ans. The body neither gains nor loses energy from the heat itself: drinking something hot adds a little heat energy, but the body must spend energy to cool it to body temperature, and eating hot food does not add chemical energy. The energy we gain comes from the chemical energy in the food itself, not its temperature.

Q4. How is carbon dioxide important for photosynthesis?

Ans. Carbon dioxide is essential for photosynthesis because it is the carbon source from which all sugars and organic molecules are built. In the Calvin cycle, CO_2 is fixed to RuBP to form PGA; without

CO_2 the plant can make ATP and NADPH in the light reactions but cannot make any food.

Chapter 6: Acellular Life

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. Which statement is not true for viruses?

- These are acellular bodies
- Their capsids are made up of proteins
- These are obligate intracellular parasites
- ✓ They have protoplasm

Q2. Which of the following is single stranded RNA virus?

- Paro virus
- Pox virus
- Reo virus
- ✓ Retro virus

Q3. The number of tail fibers in a bacteriophage is:

- 4
- 14
- ✓ 6
- 16

Q4. The protein coat of a virus is called the:

- ✓ Capsid
- Envelope
- Capsomere
- Viral membrane

Q5. The viral DNA incorporated into a lysogenic cycle is called:

✓ Prophage

Latent phage

Bacteriophage

Oncogenic virus

Q6. Carbohydrate-protein complexes that project from the surface of some viruses are:

Caspid

Envelope

Capsomeres

✓ Spikes

Q7. The transmembrane in HIV is made up of:

gp 14

✓ gp 41

gp 120

gp 210

Q8. Herpes simplex virus (HSV) is:

Non enveloped single stranded DNA virus

Non-enveloped single stranded RNA virus

Enveloped single stranded DNA virus

✓ Enveloped double stranded DNA virus

Q9. The curl leaf disease is a viral disease, the virus of this disease is spread by:

House fly

✓ White fly

Drosophila fly

Tse-tse fly

Q10. 'Kuru' disease in human beings is caused by:

Viroids

Viruses

✓ Prions

Fungi

Q11. The function of integrase enzyme in HIV is:

Break down the large protein particles

Helps in uncoating of HIV

✓ Facilitates the combination of DNA of HIV with host chromosome

Helps in reverse transcription

Q12. What type of infectious agent causes potato spindle tuber disease?

Prion

✓ Viroid

Bacteria

Virus

Exercise - Section II: Short Response Questions

Q1. Define obligate parasite, capsid, prophage and induction.

Ans. An obligate parasite is an organism that can live and

reproduce only inside a living host cell. The capsid is the protein coat that surrounds and protects the viral nucleic acid. A prophage is the viral DNA that has become integrated into the host bacterium's chromosome during the lysogenic cycle. Induction is the process by which a prophage is activated, leaves the host chromosome and enters the lytic cycle.

Q2. What is an enveloped virus? How does an envelope develop around capsid?

Ans. An enveloped virus is a virus whose capsid is surrounded by a lipoprotein envelope. The envelope develops when the virus leaves the host cell: it buds through the host's cell membrane (or an internal membrane) and takes a piece of that membrane with it, studded with viral glycoproteins (spikes) that help it attach to the next host cell.

Q3. What do you know about the status of virus?

Ans. The status of viruses is debated because they show both living and non-living characters. Outside a host they are inert particles — no metabolism, no growth, and they can even be crystallised — so they seem non-living. Inside a host cell they reproduce, mutate and evolve, which are properties of living things. They are therefore described as lying on the borderline between living and non-living.

Q4. Write a comprehensive note on the parasitic nature of viruses.

Ans. Viruses are obligate intracellular parasites: they have no metabolic machinery of their own, so they cannot make energy or proteins by themselves. They must enter a living host cell and use

the host's ribosomes, enzymes and raw materials to copy their nucleic acid and build new virus particles. Outside a host they are inactive, but inside they take over the cell completely, often killing it when new viruses are released.

Q5. How do viruses survive without a host?

Ans. Strictly, viruses do not survive without a host in the sense of living or growing — outside a host they are dormant particles. However, they can persist for some time in the environment (on surfaces, in water or in droplets) because their protein capsid, and sometimes envelope, protects the nucleic acid until they reach a new host cell.

Q6. Explain with examples that viruses are host specific.

Ans. Viruses are host-specific because their surface proteins (spikes) fit only the receptors of particular host cells, like a key in a lock. For example, the tobacco mosaic virus infects tobacco plants but not animals; the polio virus infects only humans and some primates; and bacteriophages infect only specific bacteria, so a phage that attacks one bacterial species cannot infect another.

Q7. What are the opportunistic diseases that attack AIDS victims?

Ans. AIDS victims, whose immune system is destroyed by HIV, are attacked by opportunistic diseases that a healthy body would resist, for example tuberculosis, pneumonia (Pneumocystis), oral and oesophageal candidiasis (thrush), Kaposi's sarcoma, chronic diarrhoea and certain cancers.

Q8. Why are viral diseases more difficult to treat than bacterial diseases?

Ans. Viral diseases are harder to treat than bacterial diseases because viruses live inside the host's own cells and use the host's machinery, so a drug that kills the virus often harms the host cell too. Bacteria, by contrast, are independent cells with their own walls and metabolism, so antibiotics can attack them without harming human cells. Viruses also mutate rapidly, quickly becoming resistant to drugs.

Q9. What are the limitations of the vaccine to control the flu?

Ans. Flu vaccines have limitations because there is no single flu virus: many strains cause flu, and the virus changes its genetic make-up rapidly by mutation and by exchanging genes. A vaccine made for one year's strains may not match the next year's strains, so the vaccine is remade every year, and flu-like symptoms caused by allergies or bacteria are not prevented by a flu vaccine at all.

Q10. Write a note on the cotton leaf curl disease.

Ans. Cotton leaf curl disease (CLCD) is a serious viral disease of cotton, spread by the whitefly. The virus (a begomovirus) causes the leaves to curl upward, thicken and become stunted, with small leaf-like outgrowths (enations) on the underside. Infected plants grow poorly and produce far less cotton, causing heavy losses to farmers; control depends on controlling whiteflies and growing resistant varieties.

Q11. Write down the names of diseases that are caused by prions

and viroids.

Ans. Diseases caused by prions (infectious proteins) include mad cow disease (BSE) in cattle, scrapie in sheep, and kuru and Creutzfeldt-Jakob disease in humans. Diseases caused by viroids (single circular RNA molecules) include potato spindle tuber disease, cucumber pale fruit disease and citrus exocortis.

Q12. State how viral infections cause global economic loss.

Ans. Viral infections cause global economic loss in several ways: epidemics and pandemics (such as bird flu and COVID-19) kill livestock and people, reduce the workforce and close businesses; farmers lose crops and animals (for example poultry losses from bird flu raise food prices); trade and travel are disrupted; and governments must spend billions on treatment, vaccination and control programmes.

Exercise - Section III: Extended Response Questions

Q1. Write a chronological detail about discovery of virus.

Ans. The discovery of viruses was gradual: in 1883 Adolf Mayer found that tobacco mosaic disease could be transmitted by sap even after filtering out bacteria; in 1892 Dmitri Ivanowsky showed the agent passed through a bacteria-proof filter; in 1898 Martinus Beijerinck called it a 'contagious living fluid' and named it a virus; in 1935 Wendell Stanley crystallised tobacco mosaic virus, showing it was a chemical particle; and the electron microscope (1940s) finally revealed virus particles directly.

Q2. What are the bases of classification of viruses? Explain each

type with examples.

Ans. Viruses are classified on several bases: (1) Host — plant viruses (TMV, only RNA), animal viruses (polio, herpes, HIV) and bacteriophages (only DNA, infect bacteria). (2) Nucleic acid — DNA viruses (poxvirus, herpesvirus) or RNA viruses (polio, influenza, HIV). (3) Capsid symmetry — helical (TMV), icosahedral (polio, adenovirus) or complex (bacteriophage). (4) Envelope — enveloped (HIV, influenza) or non-enveloped (polio, TMV). (5) Strands — single-stranded (retrovirus) or double-stranded (herpesvirus, reovirus).

Q3. Explain the structure of a bacteriophage.

Ans. A bacteriophage (for example T4) has a complex structure: an icosahedral head (capsid) containing double-stranded DNA; a collar and a contractile tail sheath surrounding a hollow tail core; a base plate at the end of the tail; and six tail fibres attached to the base plate. The tail fibres recognise and attach to the bacterial surface, and the sheath contracts to inject the DNA into the bacterium.

Q4. Describe the lytic cycle of a bacteriophage.

Ans. In the lytic cycle, the phage attaches to the bacterium by its tail fibres, injects its DNA, and takes over the host cell: the bacterial DNA is broken down and the host's machinery is forced to make many copies of phage DNA and phage proteins. New phage particles are assembled inside the cell, and finally the enzyme lysozyme bursts (lyses) the bacterium, releasing about 100-200 new phages to infect other bacteria.

Q5. Explain the life cycle of HIV in detail.

Ans. The life cycle of HIV: (1) Attachment — gp120 spikes bind CD4 receptors on helper T-cells. (2) Entry — the viral envelope fuses with the cell membrane and the capsid enters. (3) Reverse transcription — reverse transcriptase copies viral RNA into DNA. (4) Integration — integrase inserts the viral DNA into the host chromosome, where it may stay dormant. (5) Replication — host machinery makes viral RNA and proteins. (6) Assembly and budding — new viruses assemble, bud through the membrane (gaining an envelope), and protease matures them into infectious particles. The destruction of helper T-cells causes AIDS.

Q6. Describe the Herpes Simplex.

Ans. Herpes simplex is a highly contagious, naturally occurring disease of humans caused by herpes simplex virus (HSV), an enveloped double-stranded DNA virus. HSV-1 usually causes cold sores around the mouth and HSV-2 causes genital herpes. The virus enters through breaks in skin or mucous membranes, multiplies, and then becomes latent in nerve cells; it can reactivate later (under stress or illness) to cause new sores. There is no complete cure; antiviral drugs control the symptoms.

Q7. Explain the Polio disease in detail.

Ans. Polio (poliomyelitis) is a highly infectious viral disease caused by poliovirus, a non-enveloped RNA virus spread mainly through contaminated food and water (faecal-oral route). In most people it causes only mild illness, but in some the virus attacks motor nerve cells of the spinal cord, causing breathing problems, paralysis of

the limbs and even death. There is no cure; prevention is by vaccination (oral polio vaccine and injectable vaccine), which has almost eradicated the disease worldwide.

Q8. What are the differences between the lytic and lysogenic cycles of bacteriophage?

Ans. In the lytic cycle the phage DNA takes over the bacterium at once, makes many new phages, and bursts the cell quickly (a master-slave relationship). In the lysogenic cycle the phage DNA joins the host chromosome as a prophage and is copied quietly with the bacterium for many generations without killing it (a host-guest relationship). The prophage may later be induced (for example by UV light) to leave the chromosome and enter the lytic cycle.

Activity (model answers)

Q1. Label different components of HIV.

Ans. The components of HIV, from outside inward, are: the envelope (lipoprotein layer from the host membrane); the surface spikes gp120 (which bind CD4 on T-cells) and the transmembrane protein gp41; the matrix protein layer; the cone-shaped capsid; inside it, two strands of single-stranded RNA; and the enzymes reverse transcriptase, integrase and protease.

Q2. Write down the pathogen of each disease.

Ans. Pathogens: (1) Mad cow disease — prion. (2) Cucumber pale fruit — viroid. (3) Genital herpes — herpes simplex virus (HSV-2). (4) Kuru — prion. (5) Potato spindle tuber disease — viroid. (6) CLCD (cotton leaf curl disease) — cotton leaf curl virus (a

begomovirus, spread by whitefly).

Chapter 7: Prokaryotes

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. The percentage of water in the cytoplasm of a bacterium is:

- 30%
- 90%
- 55%
- ✓ 68%

Q2. The small rounded broken rings of DNA in a bacterial cell are known as:

- ✓ Pasminds
- Plasmodesmata
- Plasma
- Platelets

Q3. Which of the following terms describes most of the bacteria?

- Anaerobic
- Pathogens
- Many-celled
- ✓ Beneficial

Q4. What is the name for rod shaped bacteria?

- ✓ Bacilli
- Cocci
- Spirilla

Colonies

Q5. What structure allows bacteria to stick to surfaces?

✓ Pili

Flagella

Chromosome

Cell wall

Q6. The percentage of water in the cytoplasm of a bacterium is:

30%

90%

55%

✓ 68%

Q7. Which of the following terms describes most of the bacteria?

Anaerobic

Pathogens

Many-celled

✓ Beneficial

Q8. Which of these organisms are recyclers in the environment?

Producers

Carnivores

✓ Saprophytes

Pathogens

Q9. How much oxygen is produced by cyanobacteria?

✓ 21%

31%

41%

61%

Q10. In peptidoglycan, there are how many short side chains of amino acids?

1-3

✓ 4-5

6-7

9-10

Q11. The curved form of bacterium is called:

Coccus

Palisade

✓ Vibrio

Spirochete

Q12. The hollow helical structure arising from the hook in a bacterial flagellum is called:

Cilium

Plasmid

✓ Filament

Rod

Q13. The process of conjugation in bacteria was discovered in 1946 by:

Fred Griffith

J. Lederberg

E. L. Tatum

✓ Both b and c

Q14. Which of the following cannot be found in a bacterial cell?

Ribosomes

✓ Nucleus

Chromosome

Cytoplasm

Q15. A bacterium with a tuft of flagella at one side of the body:

✓ Lophotrichous

Amphitrichous

Peritrichous

None of the above

Q16. Asexual reproduction in bacteria is called:

Budding

✓ Binary Fission

Multiple fission

Both A and B

Exercise - Section II: Short Response Questions

Q1. How are chemosynthetic bacteria different from photosynthetic bacteria?

Ans. Chemosynthetic bacteria make their food using energy from chemical reactions, for example oxidising sulphur, iron or ammonia compounds, and they do not need light. Photosynthetic bacteria (such as cyanobacteria) make their food using light energy, with

pigments such as chlorophyll and bacteriochlorophyll, releasing oxygen (cyanobacteria) or not (purple sulphur bacteria).

Q2. Name any three bacteria which cause diseases in plants.
Ans. Three plant diseases caused by bacteria are: citrus canker (Xanthomonas citri), fire blight of apple and pear (Erwinia amylovora), and bacterial soft rot of potato and vegetables (Erwinia carotovora).

Q3. What are the differences between bacteria and archaea?
Ans. Bacteria (eubacteria) have peptidoglycan in their cell walls, their membrane lipids are fatty-acid esters, and their ribosomes and RNA polymerase resemble those of eukaryotes. Archaea (archaebacteria) have no peptidoglycan (they have pseudopeptidoglycan or other wall polymers), their membrane lipids are ether-linked isoprenoids, and they live in extreme environments (hot springs, salty lakes, marshes) and are genetically closer to eukaryotes in their transcription machinery.

Q4. Write short note on peptidoglycan and slimy capsule.
Ans. Peptidoglycan is the rigid mesh-like polymer that forms the bacterial cell wall; it is made of alternating NAG and NAM sugars cross-linked by short peptide chains (4-5 amino acids), and it gives the cell its shape and protection. The slimy capsule is a loose, gelatinous layer of polysaccharides outside the wall that protects the bacterium from drying and from white blood cells, and helps it stick to surfaces.

Q5. How do Gram positive bacteria differ from Gram negative bacteria?

Ans. Gram-positive bacteria have a thick peptidoglycan wall (many layers) and no outer membrane; they retain the crystal violet stain and appear purple. Gram-negative bacteria have a thin peptidoglycan layer plus an outer membrane containing lipopolysaccharide; they lose the violet stain and take the red counterstain, appearing pink. Gram-negative bacteria are often more resistant to antibiotics because of their outer membrane.

Q6. Write some harmful and useful effects of bacteria.

Ans. Harmful effects: bacteria cause diseases of humans (cholera, tuberculosis, typhoid), animals and plants, and they spoil food. Useful effects: they fix nitrogen in soil (Rhizobium), make curd, cheese and vinegar, produce antibiotics and vitamins, decompose dead matter and recycle nutrients, and are used in genetic engineering and sewage treatment.

Q7. Write a brief note on cholera.

Ans. Cholera is a serious diarrhoeal disease caused by the bacterium Vibrio cholerae, a curved (vibrio) Gram-negative rod spread by contaminated water and food. The bacteria produce a toxin in the small intestine that causes massive loss of water and salts, leading to severe dehydration, which can kill within hours if untreated. Treatment is oral rehydration salts and antibiotics; prevention is clean water and sanitation.

Q8. What do you know about the soft rot disease in plants?

Ans. Soft rot is a bacterial disease of plants, mainly caused by Erwinia (Pectobacterium) species. The bacteria produce pectin-degrading enzymes that break down the middle lamella between plant cells, so the tissue becomes soft, watery and foul-smelling and collapses. It attacks potato tubers, carrots, onions and other fleshy vegetables, especially in wet storage conditions.

Q9. Differentiate between the following terms: (i) Lytic and lysogenic cycle (ii) Lysosome and mesosome (iii) Autotrophic and heterotrophic (iv) Plasmid and bacterial DNA (v) Spherical and spiral bacteria (vi) Lag phase and log phase.

Ans. (i) Lytic cycle: phage multiplies at once and bursts the host cell. Lysogenic cycle: phage DNA joins the host chromosome as a prophage and is copied quietly with the cell. (ii) Lysosome: a eukaryotic organelle containing digestive enzymes. Mesosome: an infolding of the bacterial plasma membrane that increases surface area and helps in respiration and division. (iii) Autotrophic: organisms that make their own food from inorganic material (photosynthetic or chemosynthetic). Heterotrophic: organisms that obtain food from other organisms. (iv) Plasmid: small, circular, extra-chromosomal DNA in bacteria carrying a few genes (for example antibiotic resistance). Bacterial DNA: the main large circular chromosome carrying most genes. (v) Spherical bacteria: cocci, round cells (single, chains or clusters). Spiral bacteria: spirilla and spirochetes, twisted or corkscrew cells. (vi) Lag phase: the initial period when bacteria adjust to the medium and do not divide. Log phase: the period of rapid exponential division.

Q10. What do you know about the economic importance of bacteria?

Ans. Bacteria are economically very important: they fix nitrogen and improve soil fertility; they are used to make dairy products (curd, cheese), vinegar, alcohol and antibiotics; they decompose waste and sewage; they are used in biotechnology to produce insulin and other proteins; and their harmful side causes losses through disease and food spoilage, which also costs money for treatment and control.

Q11. What are bacterial flora? What do you know about their types?

Ans. Bacterial flora (normal flora) are the microorganisms, especially bacteria, that routinely live on and in the body of a healthy person without causing disease. Their types are: resident flora, which are always present (for example Lactobacillus in the gut and vagina, Staphylococcus on skin); and transient flora, which are present temporarily and come from the environment. They protect us by competing with pathogens, help digestion, and produce vitamins such as vitamin K and some B-vitamins.

Exercise - Section III: Extended Response Questions

Q1. Describe the structure of a typical bacterial cell.

Ans. A typical bacterial cell has: a cell wall of peptidoglycan (often with an outer slimy capsule); a plasma membrane beneath it, sometimes folded into mesosomes; cytoplasm containing 70S ribosomes, a single circular chromosome (nucleoid, not enclosed in a membrane) and small plasmids; and surface structures — flagella for movement, pili for attachment and conjugation, and sometimes

fimbriae. There is no true nucleus, no mitochondria, no endoplasmic reticulum and no other membrane-bound organelles.

Q2. Explain in detail the shapes and sizes of bacteria.

Ans. Bacteria vary in shape: cocci are spherical (single coccus, diplococcus in pairs, streptococcus in chains, staphylococcus in clusters); bacilli are straight rods (single, diplobacillus or streptobacillus); vibrios are comma-shaped curved rods; spirilla are rigid spirals; and spirochetes are flexible corkscrews. In size, most bacteria are 0.5-5 micrometres long; cocci are about 0.5-2 micrometres across, while some large bacteria reach 10 micrometres or more. They are arranged by their division pattern into pairs, chains, clusters or filaments.

Q3. Write a comprehensive note on nutrition in bacteria.

Ans. Bacteria show several modes of nutrition: photoautotrophs (cyanobacteria) use light and CO2; chemoautotrophs use chemical energy and CO2 (nitrifying bacteria); photoheterotrophs use light but need organic carbon; chemoheterotrophs (most bacteria) obtain both energy and carbon from organic matter, as saprophytes (decomposers), parasites (pathogens) or symbionts (Rhizobium fixing nitrogen in legume roots). Some bacteria are aerobic, some anaerobic, and some facultative.

Q4. Explain the process of transduction in bacteria.

Ans. Transduction is the transfer of bacterial DNA from one bacterium to another by a bacteriophage. In generalised transduction, a phage accidentally packages a piece of host DNA

instead of its own during the lytic cycle, and injects it into a new bacterium, where it may recombine with the recipient's chromosome. In specialised transduction, a prophage leaving the chromosome carries a neighbouring piece of bacterial DNA with it into the next host. Transduction is used in genetic research and spreads traits such as antibiotic resistance.

Q5. Justify with examples, the role of bacteria in research and technology.

Ans. Bacteria are central to research and technology: E. coli is the model organism for molecular biology and is used to produce human insulin, growth hormone and vaccines by genetic engineering; Agrobacterium is used to transfer genes into plants to make transgenic crops; nitrogen-fixing bacteria improve agriculture; bacteria are used in sewage treatment, oil-spill clean-up (bioremediation) and the production of antibiotics, vitamins and fermented foods.

Q6. What are the physical and chemical methods to control the harmful bacteria?

Ans. Physical methods: high temperature (dry heat 120-126 degrees C under pressure, moist heat/steam coagulating proteins), radiation (gamma rays and UV below 300 nm), membrane filtration (for heat-sensitive liquids such as antibiotics), pasteurisation (milk at 71 degrees C for 15 seconds or 62 degrees C for 32 minutes), refrigeration (10-15 degrees C), freezing (-10 to -18 degrees C) and drying (removing water so bacteria cannot grow). Chemical methods: preservatives (acids lowering pH, salt and sugar reducing

water, sodium benzoate), antiseptics on living tissue (chloroxylenol), disinfectants on surfaces (phenol, hydrogen peroxide, alcohol), and chemotherapeutic agents/antibiotics inside the body (penicillin, tetracycline, sulphonamides).

Chapter 8: Protists and Fungi

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. The multinucleated cells are called:

- Choanocytes
- ✓ Coenocytes
- Protozoans
- Choanoflagellates

Q2. Trypanosoma is transmitted by the bite of:

- House Fly
- Drosophila Fly
- White Fly
- ✓ Tse-tse Fly

Q3. All members of which of the following group are single-celled?

- Algae
- ✓ Protozoa
- Fungi
- Oomycota

Q4. Protists which absorb nutrients from dead organisms are called:

- Photoautotrophs
- Autotrophs
- ✓ Saprobes
- Heterotrophs

Q5. Parasitic Protozoans that form spores at some stage in their life belong to:

✓ Apicomplexans
Ciliates
Actinopods
Diatoms

Q6. In Actinopods the porous shells are made up of:

✓ Silica
Calcium
Chitin
Cellulose

Q7. Physarum is an example of:

✓ Slime molds
Water molds
Black molds
Diatoms

Q8. Oomycetes show close relation with fungi and their cell wall contains:

Chitin
Muramic acid
Silica
✓ Cellulose

Q9. Algae in which the body is differentiated into blade, stipe and

holdfast belong to:

- ✓ Kelps
- Euglenoids
- Golden algae
- Green Algae

Q10. Zygomycetes are also called:

- Sac fungi
- Club fungi
- Imperfect fungi
- ✓ Conjugating fungi

Q11. Penicillium belongs to:

- Deuteromycota
- Basidiomycota
- ✓ Ascomycota
- Zygomycota

Q12. Which of the following diseases in human beings is caused by Trichophyton rubrum?

- Dandruff
- ✓ Ring worm
- Athlete's foot
- Candidiasis

Q13. Histoplasmosis in human beings is caused by:

- Fusarium
- Aspergillus

✓ Mucor

Purple ergot rye

Q14. Which one of the following fungus is responsible for the smut disease in cereal plants?

Puccinia

Rhizopus

✓ Ustilago

Aspergillus

Exercise - Section II: Short Response Questions

Q1. Write down the characteristics of ciliates.

Ans. Ciliates are animal-like protists that move and feed with many short hair-like cilia covering the body. They have two kinds of nuclei — a large macronucleus that controls daily activities and a small micronucleus used in reproduction — and they reproduce asexually by fission and sexually by conjugation. Paramecium is a common example.

Q2. Write a note on dinoflagellates.

Ans. Dinoflagellates are mostly marine, single-celled protists with two flagella that make them spin as they swim. Most have a covering of cellulose plates and contain chlorophyll, so they photosynthesise; in large numbers they cause 'red tides' that colour the sea and may poison fish and shellfish.

Q3. What do you know about the reproduction in Oomycota?

Ans. Oomycota (water molds, white rusts, downy mildews)

reproduce asexually by motile biflagellate zoospores produced in sporangia, and sexually by oogamy: a male antheridium fertilises a female oogonium, forming a thick-walled oospore that survives unfavourable conditions and later germinates.

Q4. Give two characteristics of each type of protist.

Ans. Animal-like protists (protozoa): single-celled, heterotrophic, move by pseudopodia, flagella or cilia. Plant-like protists (algae): photosynthetic, with chlorophyll and cellulose walls in many. Fungus-like protists (slime molds, oomycetes): heterotrophic absorbers, with a fungus-like phase in the life cycle; slime molds are amoeba-like in one phase and spore-forming in another.

Q5. Write down the names of four fungi and antibiotics that are prepared by them.

Ans. Four fungi and the antibiotics they produce are: Penicillium notatum/chrysogenum — penicillin; Cephalosporium (Acremonium) — cephalosporins; Aspergillus species — various antibiotics; and Streptomyces is a bacterium, not a fungus, so a fourth fungal example is Penicillium griseofulvum — griseofulvin (an antifungal antibiotic).

Q6. Why is yeast useful in scientific research?

Ans. Yeast (Saccharomyces cerevisiae) is useful in research because it is a simple eukaryote: it grows fast, is easy to culture and to modify genetically, and it shares many basic cell processes (cell cycle, DNA repair, protein secretion) with human cells. It is used to study genes, to produce proteins and vaccines, and in

baking and brewing.

Q7. What are apicomplexans? What do you know about their general characters?

Ans. Apicomplexans are parasitic, spore-forming protozoans named for the apical complex of organelles at one end of the cell, which helps them penetrate host cells. Their general characters are: they are non-motile in the adult stage, they are obligate parasites, and they have complex life cycles involving sexual and asexual phases, often in two hosts. Plasmodium (malaria) is the best-known example.

Q8. Write down a note on diatoms.

Ans. Diatoms are single-celled, photosynthetic protists with a unique glass-like cell wall of silica, made of two overlapping halves (like a box and its lid). They are major producers in aquatic ecosystems — called the 'pastures of the sea' — and their silica walls accumulate on the sea floor as diatomaceous earth, used in filters and polishes.

Q9. Write a brief note on ciliates.

Ans. Ciliates (for example Paramecium) are protozoans covered with cilia used for movement and for sweeping food into the oral groove. They have a macronucleus for everyday control and a micronucleus for reproduction, a contractile vacuole for removing excess water, and they reproduce by binary fission and by conjugation.

Q10. What are edible fungi? Give some examples of edible fungi.

Ans. Edible fungi are fungi that are safe and nutritious to eat. Examples are: button mushroom (Agaricus bisporus), oyster mushroom (Pleurotus), paddy straw mushroom (Volvariella), morels (Morchella), and yeast (Saccharomyces) used in bread and brewing. Some fungi, like truffles, are prized delicacies.

Q11. What do you know about the forms of life of slime molds?

Ans. Slime molds have two forms of life: in one phase they are amoeba-like, moving and feeding as separate cells or as a multinucleate plasmodium (for example Physarum); in the other phase, when food runs out, they form a fungus-like fruiting body that produces spores, which germinate into new amoebae. They are therefore fungus-like in one phase and animal-like in another.

Q12. Explain the ecological importance of fungi.

Ans. Fungi are ecologically important as decomposers: they break down dead plants and animals with external enzymes and release essential elements (carbon, nitrogen, phosphorus) back into the soil for plants to reuse. They form mycorrhizae with plant roots, helping plants absorb water and minerals, and lichens with algae, colonising bare rock and starting soil formation. Without fungi, nutrients would stay locked in dead matter.

Exercise - Section III: Extended Response Questions

Q1. Write a detailed note on general characters of animal-like protists.

Ans. Animal-like protists (protozoa) are single-celled, eukaryotic

heterotrophs with no cell wall. They move by pseudopodia (Amoeba), flagella (Trypanosoma, Euglena), cilia (Paramecium) or by gliding; they feed by engulfing food (phagocytosis) or absorbing it, and they digest it in food vacuoles. They reproduce asexually by fission and many also have sexual reproduction (conjugation in Paramecium). They live in water, soil and as parasites; examples are Amoeba, Plasmodium and Trypanosoma.

Q2. Write a note on the general characters of Algae.

Ans. Algae are photosynthetic protists (and some plants) found in water and moist places. Their general characters are: they contain chlorophyll and other pigments and make their own food; their bodies range from single cells (diatoms, Chlamydomonas) to large multicellular seaweeds (kelps); their cell walls are usually of cellulose; they store food as starch or oils; and they reproduce asexually (by fission, spores, fragmentation) and sexually (by fusion of gametes).

Q3. Write down the general characters of Brown algae, Green algae and Red algae.

Ans. Brown algae (Phaeophyta): multicellular, large seaweeds, brown because of fucoxanthin pigment; body differentiated into holdfast, stipe and blade (kelps); store laminarin and mannitol; examples are Laminaria and Sargassum. Green algae (Chlorophyta): contain chlorophyll a and b, have cellulose walls and store starch, so they are closest to land plants; examples are Chlamydomonas, Spirogyra and Ulva. Red algae (Rhodophyta): red because of phycoerythrin, mostly marine, store floridean starch;

examples are Polysiphonia and Gelidium (source of agar).

Q4. Explain in detail the importance of Protists.

Ans. Protists are important in many ways: photosynthetic protists (diatoms, dinoflagellates, algae) produce a large share of the world's oxygen and form the base of aquatic food chains; protozoans control bacterial populations and are food for small animals; some protists cause diseases (Plasmodium causes malaria, Trypanosoma causes sleeping sickness); algae provide food, agar, alginates and iodine; and fossil protists (foraminifera, diatoms) help locate oil deposits and form diatomaceous earth.

Q5. Describe different types of fungi, giving a brief introduction of each type.

Ans. The divisions of fungi are: (1) Zygomycota (conjugating fungi) — aseptate hyphae, reproduce by zygospores; example Rhizopus (bread mold). (2) Ascomycota (sac fungi) — septate hyphae, sexual spores (ascospores) formed in sacs called asci; examples are yeast, Penicillium, Aspergillus and morels. (3) Basidiomycota (club fungi) — sexual spores (basidiospores) formed on club-shaped basidia; examples are mushrooms, puffballs, rusts and smuts. (4) Deuteromycota (imperfect fungi) — no sexual stage known; examples are Alternaria and many dermatophytes.

Q6. How does sexual reproduction occur in Basidiomycota?

Ans. In Basidiomycota, sexual reproduction begins when two compatible monokaryotic hyphae fuse (plasmogamy), forming a dikaryotic mycelium with two nuclei per cell. The dikaryotic

mycelium may form a fruiting body (mushroom). In the basidia at the tips of gills, the two nuclei fuse (karyogamy) to form a diploid nucleus, which immediately divides by meiosis into four haploid nuclei. These become four basidiospores on the outside of the basidium, which are released and germinate into new monokaryotic hyphae.

Q7. Write a comprehensive note on Lichens and Mycorrhizae.

Ans. Lichens are a symbiotic association between a fungus and an alga (or cyanobacterium): the fungus provides structure, protection and absorbs water and minerals, while the alga photosynthesises and provides food. Lichens grow on bare rock, tree bark and soil, are pioneers in soil formation, and are sensitive indicators of air pollution. Mycorrhizae are a mutualistic association between soil fungi and the roots of higher plants: the fungus increases the root's absorbing surface and supplies water and minerals (especially phosphorus), while the plant supplies the fungus with sugars.

Q8. Explain with examples the diseases that are caused by fungi.

Ans. Fungal diseases: in plants — rust of wheat (Puccinia), smut of cereals (Ustilago), late blight of potato (Phytophthora, an oomycete), and powdery mildew. In animals — ringworm of dogs and horses (Trichophyton, Microsporum), lung infection of chickens (Aspergillus), and Saprolegnia infection of fish. In humans — ringworm and athlete's foot (Trichophyton), candidiasis of mouth and vagina (Candida), dandruff (Malassezia/Microsporum furfur), ergotism from ergot of rye (Claviceps), and lung infections (Aspergillus, Histoplasma); Aspergillus flavus also produces

aflatoxin, a cancer-causing chemical in stored grains.

Chapter 9: Diversity among Plants

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. The amphibians of the plant kingdom are:

- Pteridophytes
- ✓ Bryophytes
- Gymnosperms
- Angiosperms

Q2. Liverworts are closely related to:

- Algae
- Fungi
- Lichens
- ✓ Mosses

Q3. The only positive evidence of aquatic ancestry of bryophytes is:

- Thread like protonema
- Green colour
- Some forms are still aquatic
- ✓ Ciliated sperms

Q4. A fern differs from moss in having:

- An independent gametophyte
- ✓ An independent sporophyte
- Presence of archegonia
- Swimming antherozoids

Q5. Spore bearing leaf is called:

- Sorus
- Indusium
- Ramentum
- ✓ Sporophyll

Q6. The group of flowering and seed plant is called:

- Bryophytes
- Gymnosperms
- Pteridophytes
- ✓ Angiosperms

Q7. Cycads belong to:

- Non-vascular seedless plants
- Seedless vascular plants
- ✓ Non-flowering seed plants
- Flowering plants

Q8. Needle shaped foliage leaves are found in:

- Flowering plants
- Non-vascular plants
- Pteridophytes
- ✓ Gymnosperms

Q9. Megasporophyll is the term used in gymnosperm to denote:

- ✓ Carpels
- Stamens

Leaves

Female cone

Q10. In angiosperms, the two cells around the egg are called:

✓ Synergids

Oospore

Antipodal cells

Stem cells

Q11. In which of the following features does angiosperm resemble gymnosperm?

✓ Presence of ovules

Presence of vessels

Nature of endosperm

Mode of fertilization

Q12. The largest family of Angiosperms is:

Cruciferae

✓ Leguminosae

Poaceae

Brassicaceae

Q13. Androecium whorl is made of six stamens. This statement is true for which family?

✓ Mustard family

Grass family

Pea family

Potato Family

Q14. Perianth is the characteristic feature of:

- ✓ Poaceae family
- Solanaceae family
- Mustard family
- Leguminosae family

Exercise - Section II: Short Response Questions

Q1. Write down the characteristics of amphibious plants.

Ans. Amphibious plants (bryophytes) live on land but depend on water for sexual reproduction: their male gametes (antherozoids) are flagellated and must swim through water to reach the egg in the archegonium. They are small, non-vascular, lack true roots, stems and leaves, and their main plant body is the haploid gametophyte, which bears antheridia and archegonia.

Q2. What do you know about the general characters of seedless vascular plants?

Ans. Seedless vascular plants (pteridophytes) have true roots, stems and leaves and a vascular system of xylem and phloem, so they grow taller than bryophytes. Their main plant body is the diploid sporophyte, which bears spores in sporangia on sporophylls; the gametophyte (prothallus) is small, heart-shaped and independent. They still need water for fertilisation, and they produce no seeds or flowers.

Q3. What are the differences between gametes and spores?

Ans. Gametes are haploid reproductive cells (male

sperm/antherozoid and female egg) that fuse in fertilisation to form a diploid zygote; they are produced by the gametophyte. Spores are usually haploid single cells produced by the sporophyte by meiosis; they do not fuse but germinate directly into a new gametophyte. So gametes unite to form a zygote, while spores grow alone into a new plant.

Q4. Describe the classification of angiosperms.

Ans. Angiosperms are classified into two classes: Monocotyledonae (one cotyledon, parallel leaf venation, fibrous roots, floral parts in threes, scattered vascular bundles; for example grasses, wheat, maize) and Dicotyledonae (two cotyledons, net leaf venation, tap root, floral parts in fours or fives, vascular bundles in a ring; for example pea, mustard, potato). Important families studied are Brassicaceae (mustard), Solanaceae (potato) and Poaceae (grasses).

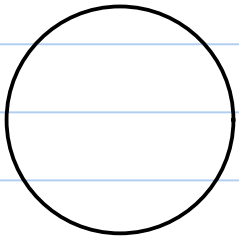
Q5. Differentiate between the following: (i) Sporophyte and gametophyte (ii) Antheridia and archegonia (iii) Male gametophyte and female gametophyte (iv) Gymnosperms and angiosperms.

Ans. (i) Sporophyte: the diploid, spore-producing generation (the main plant in ferns and seed plants). Gametophyte: the haploid, gamete-producing generation (the main plant in bryophytes). (ii) Antheridia: male sex organs producing antherozoids. Archegonia: female sex organs, flask-shaped, each containing one egg. (iii) Male gametophyte: produces male gametes (in angiosperms, the pollen tube with two sperms). Female gametophyte: produces the egg (in angiosperms, the embryo sac with egg, synergids, antipodals and

polar nuclei). (iv) Gymnosperms: naked-seeded plants, ovules not enclosed in an ovary, no flowers or fruits, endosperm haploid. Angiosperms: flowering plants, ovules enclosed in an ovary, seeds inside fruits, endosperm triploid formed by double fertilisation.

Q6. With the help of diagram describe the structure of flower.

Ans. A typical flower has four whorls on a thalamus: the calyx (sepals, usually green, protecting the bud), the corolla (petals, usually coloured, attracting insects), the androecium (stamens: filament + anther producing pollen) and the gynoecium (carpels: stigma, style and ovary containing ovules). The diagram below labels these parts: sepal, petal, stamen (anther, filament), carpel (stigma, style, ovary, ovule) and thalamus.



Flower: calyx (sepals), corolla (petals), androecium (stamen: anther + filament), gynoecium (carpel: stigma, style, ovary with ovules), thalamus

Exercise - Section III: Extended Response Questions

Q1. Explain the life cycle of Polytrichum.

Ans. Polytrichum (a moss) shows alternation of generations. The main plant is the haploid gametophyte: a leafy shoot with rhizoids, bearing antheridia (male) and archegonia (female) at the tips.

Flagellated antherozoids swim in water to the egg; fertilisation forms a diploid zygote, which grows into the sporophyte — a foot, a long seta and a capsule (sporangium) — that remains attached to and dependent on the gametophyte. Inside the capsule, meiosis produces haploid spores, which germinate into a protonema and then into a new gametophyte.

Q2. Write a detailed note on life cycle of a fern plant.

Ans. In a fern, the main plant is the diploid sporophyte: roots, an underground rhizome and large leaves (fronds) bearing sori (clusters of sporangia) on their undersides. Meiosis in the sporangia produces haploid spores, which germinate into a small, heart-shaped, independent gametophyte called the prothallus, bearing antheridia and archegonia. Antherozoids swim in water to fertilise the egg; the zygote grows into a new sporophyte, which at first depends on the prothallus and then becomes independent.

Q3. Explain the life cycle of Pinus tree in detail.

Ans. Pinus is a gymnosperm: the main plant is a tall sporophyte with needle leaves and two kinds of cones. Male cones produce pollen grains (male gametophytes); female cones bear ovules on megasporophylls. Wind carries pollen to the ovule (pollination); the pollen tube grows slowly, and after about a year one sperm fertilises the egg, forming a zygote. The zygote develops into an embryo inside the seed, which remains naked on the cone scale; the seed germinates into a new sporophyte. There are no flowers or fruits, and the endosperm is haploid.

Q4. Discuss the life cycle of flowering plants.

Ans. In flowering plants, the sporophyte bears flowers. Inside the anther, meiosis produces pollen grains (male gametophytes with a tube nucleus and two sperms). Inside the ovule, the embryo sac (female gametophyte) contains the egg, two synergids, three antipodals and a central cell with two polar nuclei. Pollination brings pollen to the stigma; the pollen tube grows to the embryo sac and double fertilisation occurs: one sperm fuses with the egg to form the diploid zygote (embryo), and the other fuses with the polar nuclei to form triploid endosperm. The ovule becomes the seed and the ovary becomes the fruit.

Q5. Write down the distribution, vegetative characters, floral characters and economic importance of family Poaceae.

Ans. Distribution: Poaceae (grass family) is distributed worldwide, in fields, lawns and wild grasslands. Vegetative characters: fibrous adventitious roots; erect, jointed stems, hollow (bamboo) or filled with stored food (sugarcane); narrow, long leaves forming a sheath around the stem. Floral characters: small flowers in dense spikes made of spikelets with two glumes, each flower enclosed in a lemma and palea; calyx and corolla are not distinguishable — the perianth is reduced to two lodicules; stamens usually three with free anthers; carpels 2-3, syncarpous, with feathery stigmas; fruit a caryopsis (grain); seeds endospermic and monocotyledonous. Economic importance: the major food source of humans and animals — wheat (Triticum aestivum), rice (Oryza sativa), maize (Zea mays), sugarcane (Saccharum officinarum, the world's major sugar source); bamboos for construction and baskets; grasses for

lawns, paper, ropes, aromatic oils and brewing (barley).

Activity (model answers)

Q1. Make a list of the biological names of all the plants given in the families of Angiosperms. Collect six different flowers and record their sepals, petals, stamens and carpels.

Ans. Plants named in the angiosperm families of this chapter include: wheat — Triticum aestivum; rice — Oryza sativa; maize (corn) — Zea mays; sugarcane — Saccharum officinarum; barley — Hordeum vulgare; lawn grass — Cynodon dactylon; bluegrass — Poa; munj — Saccharum munja. (For the flower-collection table, collect six local flowers and record, for example: rose — Rosa — sepals 5, petals 5, stamens many, carpels many; mustard — Brassica — sepals 4, petals 4, stamens 6, carpel 1; pea — Pisum sativum — sepals 5, petals 5, stamens 10, carpel 1; potato — Solanum tuberosum — sepals 5, petals 5, stamens 5, carpel 1; China rose — Hibiscus — sepals 5, petals 5, stamens many, carpels 5; wheat — Triticum aestivum — perianth as lodicules, stamens 3, carpel 1.)

Chapter 10: Form and Functions in Plants

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. Which one of the following is not micronutrient?

- Mn
- Fe
- ✓ Mg
- Zn

Q2. Which one of the following is macronutrient?

- Cu
- Mo
- B
- ✓ K

Q3. Which of the following is an essential element of organic compounds?

- Manganese
- Nitrogen
- ✓ Carbon
- Sodium

Q4. Which one of following is the micronutrient?

- Oxygen
- Calcium
- ✓ Copper

Nitrogen

Q5. Apoplast is the movement of water through:

✓ Interspaces of cell walls

Chloroplast

Vacuole

Protoplasm

Q6. Plants which live in fresh water are called:

✓ Hydrophytes

Xerophytes

Mesophytes

Halophytes

Q7. Carnivorous plants use insects as source of:

Water

Glucose

Oxygen

✓ Nitrogen

Q8. Thigmotropism is an example of:

Nastic movement

Tactic Movement

✓ Tropic Movement

None of these

Q9. If proper cold periods are not available to perennial plants like tulips and daffodils then:

They may develop large flowers

It results in increased seed production

✓ It may result in delayed or failure to flower

They produce more underground storage organs

Q10. The apoplast pathway is discontinuous in the endodermis due to:

Movement of water and minerals into xylem

Presence of plasmodesmata

✓ Presence of casparian strips

Lack of water in cortex

Q11. The pH in guard cells decreases during the night because:

Photosynthesis increases CO_2 level

✓ No photosynthesis occurs, increasing CO_2 concentration

Starch is broken down into glucose

Water enters the guard cells

Q12. The scientific name of bladderwort is:

Drosera

✓ Utricularia

Dionaea

Nepenthes

Q13. Spongy mesophyll tissue is located:

Just below the upper epidermis

Just above the lower epidermis

✓ Under the palisade and above the lower epidermis

Above the palisade and below the upper epidermis

Q14. According to Levitt's hypothesis, the opening of stomata occurs due to:

Increase in temperature

✓ Movement of K+ ions into guard cells

High concentration of CO_2 inside leaves

Accumulation of malate in epidermal cells

Q15. The 'TACT' theory for the ascent of sap was presented in 1894 by:

Lysenko

✓ Dixon and Jolly

W. M. Stanley

Fredrick Griffith

Q16. The opening of stomata depends upon:

Influx of Na+

Influx of Ca++

✓ Influx of K+

Influx of Mg++

Q17. Which mechanism was presented by E. Munch in 1927?

Apoplast pathway

Vacuolar pathway

✓ Pressure flow

Plasmolysis

Q18. In which condition is water transported into the cell from outside, when the solution inside the cell is:

- ✓ Hypertonic
- Hypotonic
- None of these
- Isotonic

Q19. Vernalization process was first of all used in 1920 by:

- E. Munch
- Darwin
- Dixon
- ✓ Lysenko

Exercise - Section II: Short Response Questions

Q1. Enlist the inorganic plant nutrients.

Ans. Inorganic plant nutrients are the mineral elements plants absorb from the soil. Macronutrients (needed in large amounts): nitrogen, phosphorus, potassium, calcium, magnesium, sulphur. Micronutrients (needed in small amounts): iron, manganese, zinc, copper, boron, molybdenum, chlorine and nickel.

Q2. Write a note on insectivorous plants.

Ans. Insectivorous (carnivorous) plants grow in nitrogen-deficient soils (bogs, marshes) and supplement their nitrogen by trapping and digesting insects. Examples: pitcher plant (Nepenthes) with a pitfall trap; Venus flytrap (Dionaea) with snap traps; sundew (Drosera) with sticky tentacles; bladderwort (Utricularia) with suction bladders. They still photosynthesise; insects provide mainly

nitrogen.

Q3. How do mesophyll cells help in gaseous exchange?

Ans. Mesophyll cells help gaseous exchange because the leaf interior has large air spaces between the loosely packed spongy mesophyll cells. Carbon dioxide from the air enters through the stomata, diffuses through these air spaces and dissolves in the moist cell walls, reaching every photosynthetic cell; oxygen produced in photosynthesis leaves by the same route.

Q4. Write a note on annual rings.

Ans. Annual rings are the concentric rings seen in the wood of a tree trunk. Each year, the vascular cambium produces spring wood (large, thin-walled vessels, light in colour) in the growing season and autumn wood (small, thick-walled vessels, dark) later; one light + one dark band make one annual ring. Counting the rings gives the age of the tree, and their width records the climate of each year.

Q5. What is the role of stomata in gaseous exchange?

Ans. Stomata are the pores in the leaf epidermis, each guarded by two guard cells. They are the main route for gaseous exchange: carbon dioxide enters through them for photosynthesis, and oxygen and water vapour leave through them. By opening and closing (controlled by turgor changes in the guard cells), stomata balance the plant's need for CO_2 against water loss by transpiration.

Q6. Transpiration and gaseous exchange are interconnected

processes, discuss.

Ans. Transpiration and gaseous exchange are interconnected because both occur through the same stomata: when stomata open to let CO2 in for photosynthesis, water vapour escapes at the same time (transpiration). The plant cannot take in CO2 without losing water, and it cannot reduce water loss by closing stomata without starving photosynthesis. This trade-off controls the plant's water balance and also drives the ascent of sap.

Q7. What is the pressure potential?

Ans. Pressure potential (denoted by the symbol Psi-p) is the pressure exerted by the protoplast on the cell wall from inside — in other words, the turgor pressure of the cell. It is usually positive in a turgid cell and pushes water out of the cell; water potential is the sum of solute potential (negative) and pressure potential.

Q8. Differentiate between apoplast and symplast pathway for the movement of water.

Ans. In the apoplast pathway, water moves through the continuous system of cell walls and intercellular spaces, without crossing any plasma membrane, until it reaches the endodermis. In the symplast pathway, water moves from cell to cell through the cytoplasm, passing through plasmodesmata, so it crosses plasma membranes. The apoplast is faster but is blocked at the Casparian strips of the endodermis, forcing water into the symplast.

Q9. What do you know about osmotic adjustment in plants?

Ans. Osmotic adjustment is the physiological process by which

plants regulate the solute concentration inside their cells to maintain osmotic potential and water balance. Under drought or salt stress, the plant accumulates solutes (sugars, amino acids such as proline, ions) in the cell, lowering its solute potential so that water continues to enter the cell and turgor is maintained.

Q10. Differentiate among Hydrophytes, Xerophytes, Mesophytes and Halophytes.

Ans. Hydrophytes live in water or very wet places (for example water lily, Hydrilla); they have large air spaces, reduced roots and thin cuticles. Xerophytes live in dry places (cactus, Acacia); they have thick cuticles, sunken stomata, reduced leaves or spines, and water-storing tissue. Mesophytes live in moderate conditions (most crop plants); they have ordinary leaves and roots and need regular water. Halophytes live in salty soils (mangroves); they tolerate or excrete salt and often have fleshy leaves and breathing roots.

Q11. What are the differences between source and sink?

Ans. The source is the plant part that produces or releases food, usually a mature photosynthesizing leaf (or a storage organ when it is being emptied). The sink is the part that receives and uses or stores food, such as growing roots, shoot tips, fruits, seeds and storage organs. Food moves in the phloem from source to sink by the pressure-flow mechanism, and an organ can change from sink to source (for example a leaf in spring) with the season.

Q12. What is growth? Differentiate between the primary and secondary growth.

Ans. Growth is the permanent increase in size, weight, form and volume of an organism over time, brought about by cell division, cell enlargement and cell differentiation. Primary growth is the increase in length of roots and shoots by division of apical meristems. Secondary growth is the increase in thickness or diameter by division of lateral meristems (vascular cambium and cork cambium), producing wood and bark.

Q13. Define tactic, nastic and tropic movements.

Ans. Tactic movements are movements of the whole organism towards or away from a stimulus (for example a motile alga swimming towards light — phototactic). Nastic movements are non-directional movements of a plant part in response to a stimulus, where the direction is fixed by the plant, not the stimulus (for example the folding of Mimosa leaves on touch — thigmonastic — and the opening of flowers in light). Tropic movements are directional growth movements of a plant part towards or away from a stimulus (for example bending towards light — phototropism, roots growing towards gravity — geotropism, tendrils coiling around support — thigmotropism).

Q14. Is it possible that a winter variety of plant can be grown in spring by chilling treatment?

Ans. Yes, it is possible: this is vernalization. A winter variety needs a period of low temperature (chilling treatment) to become able to flower. If seeds or seedlings are given the required cold treatment artificially, a winter variety can be grown in spring and will flower in the same season instead of waiting for winter.

Exercise - Section III: Extended Response Questions

Q1. Discuss water potential in detail.

Ans. Water potential (Psi) is the total kinetic energy of water molecules in a system that determines their movement from one place to another; pure water at standard conditions has a water potential of zero, and all solutions have negative values. It is the sum of solute potential (Psi-s, always negative, lowered by dissolved solutes) and pressure potential (Psi-p, positive turgor pressure inside cells, zero or negative in xylem). Water always moves from higher (less negative) to lower (more negative) water potential, through osmosis and along the soil-plant-air continuum from root to leaf.

Q2. Write a detailed note on food conducting tissues of plants.

Ans. Food-conducting tissue in plants is the phloem. It consists of sieve tube elements (living cells without nuclei, joined end to end into sieve tubes), companion cells (which support the sieve tubes and control loading), phloem parenchyma (storage) and phloem fibres (support). Sugars made in the leaves (source) are loaded into the sieve tubes, water follows by osmosis, and the resulting pressure pushes the sap towards sinks (roots, fruits, growing tips) by the pressure-flow mechanism.

Q3. Explain the two hypotheses about the mechanism of opening and closing of stomata.

Ans. Two hypotheses explain stomatal opening and closing. (1) The starch-sugar hypothesis: in light, guard-cell starch is converted to

sugar, raising solute concentration, so water enters, guard cells become turgid and the pore opens; at night the reverse happens and the pore closes. (2) The potassium-ion (K+) hypothesis of Levitt: in light, K+ ions are actively pumped into guard cells, water follows by osmosis, the guard cells swell and the stoma opens; in darkness K+ leaves the guard cells, water follows out, and the stoma closes. The K+ hypothesis is now generally accepted.

Q4. Explain ascent of sap with the help of 'TACT' theory.

Ans. The ascent of sap is explained by the TACT theory (transpiration pull, cohesion, adhesion and tension), proposed by Dixon and Joly in 1894. Transpiration from leaves creates a pull (tension) in the xylem. Cohesion — the attraction between water molecules — holds the water column together, and adhesion — attraction between water and xylem walls — keeps it attached to the walls. Together, the continuous water column is pulled upward from roots to leaves against gravity, without any pumping by the plant.

Q5. Write a detailed note on the pressure flow mechanism for the transportation of organic food in a plant body.

Ans. The pressure-flow mechanism (Munch, 1927) explains translocation in phloem: sugars are actively loaded into sieve tubes at the source, lowering the water potential there, so water enters by osmosis and builds up a high pressure. At the sink, sugars are unloaded and water leaves, so the pressure is low. Sap therefore flows by mass flow from the high-pressure source end to the low-pressure sink end. When the source and sink swap (for example

in spring, when stored food moves from roots to buds), the direction of flow reverses.

Q6. Explain osmoregulation in plants.

Ans. Osmoregulation in plants is the control of water and solute balance to keep cells turgid and functioning. Plants adjust their internal solute concentration (osmotic adjustment) by accumulating sugars, amino acids and ions, so water keeps entering the cells even in dry or salty soil. Guard cells regulate water loss by opening and closing stomata, and roots exclude or excrete excess salt in halophytes. If osmoregulation fails, cells lose turgor, wilt and die.

Q7. How is support provided to herbaceous and woody plants?

Ans. Support in herbaceous plants comes mainly from turgor pressure: water-filled cells press against their walls and keep the soft stem upright, together with collenchyma (living cells with thickened corners) in young stems. Support in woody plants comes from secondary growth: the vascular cambium produces wood (secondary xylem) with thick lignified walls and fibres, and bark outside; this rigid skeleton supports the tree for many years.

Q8. Explain photoperiodism in detail.

Ans. Photoperiodism is the response of plants to the relative lengths of day and night, which controls flowering. Short-day plants (for example Chrysanthemum) flower when nights are longer than a critical length; long-day plants (for example spinach) flower when nights are shorter; day-neutral plants (tomato) flower regardless of day length. The pigment phytochrome detects light: red light

converts Pr to Pfr, and far-red light converts it back; the amount of Pfr at the end of the night determines whether the flowering hormone is produced.

Q9. Explain the role of different types of plant hormones in detail.
Ans. The main plant hormones are: auxins — promote cell elongation, apical dominance and root formation, and are used in rooting powders; gibberellins — promote stem elongation, seed germination and flowering, and break dormancy; cytokinins — promote cell division and delay leaf ageing; abscisic acid (ABA) — promotes dormancy and closes stomata under drought stress; and ethylene — a gas that promotes fruit ripening and leaf fall. Together they control growth, development and responses to the environment.

Q10. What are meristematic tissues? Explain the role of apical meristem in plant growth.
Ans. Meristematic tissues are groups of young, actively dividing, undifferentiated cells found at growing points. The apical meristem sits at the tips of roots and shoots: its division produces new cells that elongate and differentiate, causing primary growth — increase in length. The root apical meristem is protected by the root cap, and the shoot apical meristem also produces leaf and flower primordia. Without apical meristems, the plant could not grow taller or produce new organs.

Q11. What is phytochrome? What is the role of phytochrome in the flowering of plants?

Ans. Phytochrome is a blue-green light-sensitive pigment found in plants, existing in two forms: Pr, which absorbs red light (660 nm), and Pfr, which absorbs far-red light (730 nm). Red light converts Pr to Pfr, and far-red light (or darkness) converts Pfr back to Pr. Pfr is the active form: in short-day plants, flowering is promoted when Pfr is low at the end of a long night, while in long-day plants flowering is promoted when Pfr remains high after a short night. Phytochrome also controls seed germination and shade avoidance.

Q12. What is thermoregulation? What are the effects of low and high temperature on plants? How do plants maintain thermoregulation?

Ans. Thermoregulation in plants is the maintenance of a suitable internal temperature despite changes in the surroundings. Low temperature slows metabolism, can freeze cell water and damage membranes, and (through vernalization) is needed by biennials and winter varieties to flower. High temperature increases transpiration and respiration, denatures enzymes and can scorch leaves. Plants maintain temperature by transpirational cooling (evaporation of water from leaves), by leaf orientation and rolling, by reflective hairs and waxy cuticles, and by heat-shock proteins that protect enzymes at high temperatures.

Chapter 11: Biodiversity of Animals

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. The simplest animals belong to:

- Cnidaria
- ✓ Porifera
- Platyhelminthes
- Arthropoda

Q2. Water vascular system is found in:

- Mollusca
- Porifera
- ✓ Echinodermata
- Annelida

Q3. Acoelomates are characterized by:

- The absence of brain
- The absence of mesoderm
- ✓ A solid body without a cavity surrounding internal organs
- A coelom that is not completely lined with mesoderm

Q4. Gastrovascular cavity and nematocyst are present in:

- Porifera
- Annelida
- Nematoda
- ✓ Coelenterata

Q5. All monotreme species are found in:

- ✓ Australia
- Asia
- Europe
- Africa

Q6. Gemmules are present in poriferans which are helpful in:

- Digestion
- Sexual reproduction
- Secretion of spicules
- ✓ Survival in drought

Q7. Nematocysts are the organs of:

- Sensation
- Reproduction
- ✓ Defence and offence
- Respiration

Q8. Centipedes and millipedes belong to:

- ✓ Arthropoda
- Platyhelminthes
- Annelida
- Nematoda

Q9. Ascaris (a nematode) normally inhabits the lumen of:

- Stomach
- ✓ Small intestine

Appendix

Large intestine

Q10. The egg laying mammals are placed in sub class:

✓ Prototheria

Metatheria

Eutheria

Cyclostomata

Q11. Identify the mismatch pair in the following:

✓ Arthropoda-Acoelomata

Annelida-Coelomata

Nematoda-Pseudocoelomata

Porifera-Sponges

Q12. Salamander belongs to class:

Reptilia

Mammalia

Aves

✓ Amphibia

Q13. Radial symmetry is recorded in:

✓ Coelenterata

Platyhelminthes

Arthropoda

Mollusca

Q14. Digestive system in triploblastic animals is:

Sac like

Open

✓ Tube like

Closed

Q15. The body of sponges is supported by:

Notochord

✓ Spicules

Cartilage

Chitin

Q16. In acoelomate animals excretory system consists of:

✓ Protonephridia

Metanephridia

Malpighian tubules

Diffusion

Q17. Animals which cannot be divided into two equal halves by any plane of symmetry are called:

✓ Asymmetrical

Bilaterally symmetrical

Radially symmetrical

Biradial symmetrical

Q18. Organisms lacking body cavity are called:

✓ Acoelomates

Pseudocoelomates

Coelomate

Biradial symmetrical

Q19. Body of sponges contains one outgoing large opening called:

Ostium

✓ Osculum

Porin

Mouth

Q20. The rasping tongue is present in:

Echinoderms

Annelids

Arthropods

✓ Molluscs

Exercise - Section II: Short Response Questions

Q1. How useful is the study of the nature of body cavity and coelom in the classification of animals?

Ans. The nature of the body cavity (coelom) is a fundamental basis of animal classification: it separates animals into acoelomates (no cavity, for example flatworms), pseudocoelomates (a cavity not fully lined by mesoderm, for example roundworms) and coelomates (a true cavity fully lined by mesoderm, for example annelids, arthropods, molluscs and chordates). This grouping reflects evolutionary relationships and is linked to the complexity of organs, circulation and movement, so it helps place an animal in its correct phylum.

Q2. Distinguish between incomplete and complete digestive

system.

Ans. In an incomplete digestive system there is a single opening (mouth) that takes in food and expels waste, and digestion occurs in a sac-like gastrovascular cavity (for example in cnidarians and flatworms). In a complete digestive system there are two openings — mouth and anus — and food passes one way through a tube-like gut, allowing stepwise digestion and absorption (for example in annelids, arthropods and vertebrates).

Q3. What is the difference between direct and indirect development?

Ans. In direct development, the young animal hatches or is born looking like a small adult and simply grows larger (for example in mammals, birds and reptiles). In indirect development, the young (larva) looks very different from the adult and passes through metamorphosis to become an adult (for example the tadpole of a frog, or the caterpillar of a butterfly).

Q4. What are the peculiar features that you find in parasitic Platyhelminthes?

Ans. Parasitic flatworms (tapeworms, liver flukes) show special features for parasitic life: they have no digestive system of their own and absorb digested food through their body surface; they bear suckers and hooks for attachment; their body is covered by a resistant tegument; they are hermaphrodite and produce enormous numbers of eggs; and their life cycles often involve two hosts, with larval stages adapted for transfer between hosts.

Q5. What are the reasons that you can think of for the arthropods to constitute the largest group of the animal kingdom?

Ans. Arthropods are the largest group of the animal kingdom because: they have a light, strong, jointed exoskeleton of chitin that protects them and allows varied movement; their small size lets them occupy many niches; they reproduce rapidly and in large numbers; they can fly (insects), which aids dispersal; and they show great adaptability in feeding (herbivores, predators, parasites, scavengers) and in habitats (land, water, air).

Q6. 'All vertebrates are chordates but all chordates are not vertebrates.' Justify the statement.

Ans. All vertebrates are chordates because they possess, at some stage, the chordate characters: a notochord, a dorsal hollow nerve cord, pharyngeal gill slits and a post-anal tail. In vertebrates the notochord is replaced by the vertebral column in the adult. But all chordates are not vertebrates: some chordates, such as lancelets (cephalochordates) and tunicates (urochordates), keep the notochord throughout life or only in the larval stage and never develop a vertebral column or a true skull.

Q7. How are the presence of swim bladder, gills and fins in Pisces beneficial for aquatic life?

Ans. In fishes, gills extract dissolved oxygen from water for respiration; fins (paired and unpaired) provide balance, steering and propulsion for swimming; and the swim bladder (in bony fishes) is a gas-filled sac that controls buoyancy, letting the fish stay at a chosen depth without spending energy. Together these adaptations

make fishes fully suited to life in water.

Q8. What are the modifications that are observed in birds that help them fly?

Ans. Birds show many flight modifications: a streamlined body; forelimbs modified into wings with feathers; hollow, pneumatic bones that make the skeleton light; a keeled sternum for strong flight muscles; a high metabolic rate with a four-chambered heart and efficient lungs with air sacs; a horny beak instead of heavy teeth; and the absence of a urinary bladder, with nitrogenous waste excreted as semi-solid uric acid.

Q9. What is madreporite?

Ans. The madreporite is a small, sieve-like, calcareous plate on the upper (aboral) surface of echinoderms (starfish, sea urchins). It is the opening of the water vascular system: sea water enters through it into the stone canal and ring canal, and from there into the radial canals and tube feet, which the animal uses for locomotion, feeding and respiration.

Q10. What is metamerism?

Ans. Metamerism is the division of an animal's body into a series of similar, repeating segments (metameres), each containing similar sets of organs. It is characteristic of annelids (earthworm, leech), where the body is divided externally by grooves and internally by septa, and it is also seen in arthropods and in the early development of vertebrates.

Exercise - Section III: Extended Response Questions

Q1. Explain general characteristics of sponges.

Ans. General characters of sponges (Porifera): they are the simplest multicellular animals, mostly marine and sessile, with an asymmetrical or radially symmetrical body. They have no true tissues, organs or nervous system; the body wall has many pores (ostia) through which water enters, and one large opening (osculum) through which it leaves. The body is supported by spicules (of calcium carbonate or silica) and/or spongin fibres. Special collar cells (choanocytes) create the water current and capture food particles; digestion is intracellular. They reproduce asexually by budding, fragmentation and gemmules, and sexually by gametes; many are hermaphrodite.

Q2. Define the term acoelomate. Describe characteristics of any acoelomate worms.

Ans. An acoelomate is an animal with no body cavity (coelom) between the gut and the body wall; the space is filled with mesenchyme. Acoelomate worms (Platyhelminthes, flatworms) are triploblastic, bilaterally symmetrical, dorsoventrally flattened animals with an incomplete digestive system (a single opening, or none in tapeworms), a simple nervous system with a ladder-like arrangement, flame cells (protonephridia) for excretion, and no circulatory or respiratory system. Examples: Planaria (free-living), liver fluke and tapeworm (parasitic).

Q3. Explain general characteristics of coelomate segmented worms.

Ans. Coelomate segmented worms (Annelida) are triploblastic, bilaterally symmetrical animals with a true coelom and a body divided into similar segments (metamerism). They have a complete, tube-like digestive system, a closed circulatory system with blood vessels, excretion by nephridia (metanephridia), a ventral nerve cord with a brain-like ganglion, and locomotion by setae or parapodia. Examples: earthworm (terrestrial), Nereis (marine) and leech (parasitic, freshwater).

Q4. Write a note on spiny skinned invertebrates.

Ans. Spiny-skinned invertebrates (Echinodermata) are exclusively marine animals with a spiny skin (from the Greek 'echinos' = spiny, 'dermos' = skin) and a calcareous endoskeleton of plates. Adults are radially symmetrical (pentamerous), but larvae are bilateral. They possess a unique water vascular system with tube feet, used for locomotion, feeding and respiration, opening through the madreporite. They have no head, no excretory organs and a simple nervous system. Examples: starfish, sea urchin, sea cucumber and brittle star.

Q5. Define homeotherms and write general characteristics of birds.

Ans. Homeotherms (warm-blooded animals) keep their body temperature constant regardless of the surrounding temperature, using metabolic heat and insulation. General characters of birds (Aves): they are feathered, warm-blooded (homeotherm) vertebrates with forelimbs modified into wings; they have a horny beak, no teeth, hollow pneumatic bones, a keeled sternum, a four-chambered heart, lungs with air sacs, and a syrinx (voice box)

for sound. They lay shelled eggs, and most can fly; examples are pigeon, sparrow and ostrich.

Q6. Explain general characteristics of mammals.

Ans. General characters of mammals (Mammalia): they are warm-blooded (homeotherm) vertebrates with hair on the body and mammary glands that produce milk for the young. They have a four-chambered heart, a diaphragm separating thorax and abdomen, a well-developed brain with a large cerebrum, and external ears (pinnae). They are mostly viviparous (giving birth to young), except the egg-laying monotremes (platypus, echidna); most have teeth differentiated into incisors, canines, premolars and molars. Examples: humans, whales, bats and dogs.

Q7. Explain general characteristics of Reptilia.

Ans. General characters of Reptilia: they are cold-blooded (poikilotherm) vertebrates with dry, scaly skin that prevents water loss. They breathe by lungs, have a three-chambered heart (except crocodiles, which have four chambers), and lay shelled, amniotic eggs on land, so they do not need water for reproduction. Most have four limbs with claws (snakes have lost them), and fertilisation is internal. Examples: lizard, snake, turtle and crocodile.

Q8. What are arthropods and describe economic importance of arthropods?

Ans. Arthropods are animals with jointed legs (from the Greek 'arthros' = joint, 'podos' = leg), a segmented body, and a chitinous exoskeleton that is shed (moulted) as they grow. Their economic

importance: they are useful as pollinators (honeybee), producers of honey, silk (silkworm) and lac; they are food for many animals and humans; but they are also harmful as crop pests (locusts, aphids), as vectors of disease (mosquitoes spread malaria, houseflies spread cholera) and as stored-grain pests. Crustaceans such as prawns and crabs are important food.

Q9. Explain general characteristics of Mollusca.

Ans. General characters of Mollusca: they are soft-bodied, triploblastic, bilaterally symmetrical coelomates (the name comes from the Latin 'molluscus' = soft). The body is divided into head, muscular foot and visceral mass, usually covered by a mantle that secretes a calcareous shell. Most have a rasping tongue (radula) for feeding, a complete digestive system, an open circulatory system (except cephalopods, which have a closed system), and gills or a lung for respiration. Examples: snail, slug, oyster, octopus and squid.

Q10. Define metamerism and explain general characteristics of annelids.

Ans. Metamerism is the division of the body into a series of similar segments (metameres). General characters of annelids: triploblastic, bilaterally symmetrical, coelomate worms with a segmented body, each segment bearing setae or parapodia for movement. They have a complete digestive system, a closed circulatory system, metanephridia for excretion, a ventral nerve cord, and they reproduce sexually (earthworms are hermaphrodite). Examples: earthworm, Nereis and leech.

Q11. Differentiate between nematodes and Platyhelminthes.

Ans. Nematodes (roundworms) are unsegmented, cylindrical worms with a pseudocoelom (a body cavity not fully lined by mesoderm), a complete digestive system with mouth and anus, a tough cuticle that is moulted, and separate sexes. Flatworms (Platyhelminthes) are dorsoventrally flattened, acoelomate worms with an incomplete digestive system (a single opening or none), no anus, no circulatory system, and flame cells for excretion; many are hermaphrodite. So the main differences are: body shape (round vs flat), body cavity (pseudocoelom vs none), digestive system (complete vs incomplete) and cuticle (present in nematodes, absent in flatworms).

Activity (model answers)

Q1. Write the names of the arthropods pictured.

Ans. The arthropods pictured are (top row, left to right): a hoverfly/bee (Hymenoptera/Diptera), a ladybird beetle (Coleoptera), a butterfly (Lepidoptera), a leafhopper (Hemiptera); (bottom row, left to right): a dragonfly (Odonata), a dung beetle (Coleoptera), a crab (Crustacea) and a spider (Arachnida). Note: the spider is an arachnid, not an insect — it has eight legs and two body regions, while the insects have six legs and three body regions.

Chapter 12: Chromosome and DNA

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. Morphological characteristics of chromosomes are collectively called:

- Holotype
- Karyokinesis
- ✓ Karyotype
- Neotype

Q2. Mongolism is also known as:

- ✓ Down's syndrome
- Klinefelter's syndrome
- Turner's syndrome
- Jacob's syndrome

Q3. All are related to Turner's syndrome, except:

- Short stature
- Webbed neck
- ✓ Broad face
- Without ovaries

Q4. If a DNA molecule has 20% adenine, what is the percentage of guanine?

- 20%
- ✓ 30%

40%

60%

Q5. Synthesis of a new DNA strand begins with:

a DNA primer

✓ RNA primer

an Okazaki fragment

a DNA polymerase I

Q6. The strand that runs away from replication fork is:

leading strand

✓ lagging strand

continuous (strand)

primer

Q7. The number of chromosomes in Monkey is:

46

✓ 48

50

44

Q8. The strand of DNA that elongates towards the replication fork is:

Lagging

✓ Leading

Primer

Okazaki

Q9. The number of tRNA in human are:

45

60

✓ 61

35

Q10. Anticodons are present on:

mRNA

✓ tRNA

DNA

rRNA

Q11. The initiation codon is:

UUA

ACC

✓ AUG

GCG

Q12. Down's syndrome is autosomal non-disjunction of chromosome number:

✓ 21

22

13

18

Q13. Which condition appears due to point mutation?

Turner's syndrome

Klinefelter's syndrome

✓ Sickle cell anaemia

Down's syndrome

Q14. What is the role of histone proteins in DNA structure?

Replicates DNA

Unwind DNA

✓ Package DNA into chromatin

Synthesize RNA

Q15. Which of the following is a function of noncoding DNA?

Causes mutation

Has no function in the cell

✓ May regulate gene expression

Codes enzymes

Q16. Which strand of DNA acts as template strand in RNA transcription?

Coding strand

✓ Non-coding strand

Sense strand

Complementary strand

Q17. Which of the following is not a component of DNA nucleotide?

Nitrogenous base

Phosphate group

✓ Ribose sugar

Deoxyribose sugar

Q18. The sickle cell anaemia and phenylketonuria are caused by:

- Turner's syndrome
- Klinefelter's syndrome
- ✓ Point mutation
- Down's syndrome

Q19. Which of the following statements about prokaryotes is true?

- It is linear and found in nucleus
- It has intron and exon
- ✓ It is circular and lacks histone
- Circular and possesses histone

Q20. In eukaryotic cells mRNA is synthesized by:

- RNA poly-I
- ✓ RNA poly-II
- RNA poly-III
- primase

Exercise - Section II: Short Response Questions

Q1. Differentiate between heterochromatin and euchromatin.

Ans. Heterochromatin is highly condensed, darkly staining chromatin that remains condensed even between divisions; its genes are generally not expressed (inactive). Euchromatin is loosely packed, lightly staining chromatin that uncoils after division; its genes are active and are transcribed. Heterochromatin is mostly found near centromeres and telomeres, while euchromatin forms most of the chromosome arms.

Q2. What do you know about mechanism of positive control of gene regulation?

Ans. In positive control of gene regulation, a regulatory protein (an activator) binds to the DNA near the gene and helps RNA polymerase to start transcription, switching the gene on. For example, in the lac operon, the CAP-cAMP activator complex binds upstream of the promoter when glucose is absent, increasing transcription of the lactose-digesting genes. Without the activator, transcription remains low.

Q3. How many types of chromosome are there on the basis of centromere position?

Ans. On the basis of centromere position there are four types of chromosomes: metacentric (centromere in the middle, two equal arms), submetacentric (centromere slightly off-centre, one arm shorter), acrocentric (centromere near one end, one very short arm) and telocentric (centromere at the very end, only one arm).

Q4. Differentiate between conservative model and dispersive model of DNA replication.

Ans. In the conservative model of DNA replication, the original double helix stays completely intact and a wholly new double helix is made, so one daughter molecule is entirely old and the other entirely new. In the dispersive model, the original DNA is broken into pieces and the new DNA is mixed with old pieces in both daughter molecules, so each daughter strand contains a mixture of old and new DNA. (The Meselson-Stahl experiment proved that replication is actually semi-conservative, not either of these.)

Q5. Define RNA primer, primase and ligase.

Ans. An RNA primer is a short stretch of RNA (about 5-10 nucleotides) that provides a free 3'-OH end for DNA polymerase to start adding DNA nucleotides, since DNA polymerase cannot start a new strand on its own. Primase is the enzyme that synthesises the RNA primer. Ligase is the enzyme that joins DNA fragments (Okazaki fragments on the lagging strand, and nicks in the backbone) by forming phosphodiester bonds, sealing the new strand.

Q6. How does leading strand differ from lagging strand?

Ans. The leading strand is synthesised continuously in the 5' to 3' direction towards the replication fork, needing only one RNA primer. The lagging strand is synthesised discontinuously away from the fork, in short Okazaki fragments, each needing its own primer; the fragments are later joined by DNA ligase. The leading strand is therefore made as one long piece, while the lagging strand is made in pieces.

Q7. Write any three characteristics of codons.

Ans. Three characteristics of codons are: (1) each codon is a sequence of three bases (a triplet) on mRNA that codes for one amino acid; (2) the code is specific (one codon codes for only one amino acid) but degenerate (several codons can code for the same amino acid); (3) the code is universal (nearly the same in all organisms) and non-overlapping, and it includes a start codon (AUG) and three stop codons (UAA, UAG, UGA) that do not code for

any amino acid.

Q9. Write differences between physical and chemical mutagens.

Ans. Physical mutagens are forms of energy, such as ultraviolet light, X-rays and gamma rays, which damage DNA by breaking bonds or forming abnormal links (for example thymine dimers). Chemical mutagens are substances, such as nitrous acid, mustard gas, benzene and some dyes, which change the chemical structure of bases (for example by deamination or by inserting between bases). Physical mutagens act from outside the cell as radiation, while chemical mutagens act by reacting chemically with DNA.

Q10. Write useful aspects of mutation.

Ans. Useful aspects of mutation: mutations are the raw material of evolution — they create new alleles and new variations on which natural selection acts; they have produced useful varieties in agriculture (for example high-yielding and disease-resistant crops by induced mutation); and they are used in research and medicine (for example to study gene function and to produce new strains of microorganisms).

Q11. Write harmful aspects of mutation.

Ans. Harmful aspects of mutation: most mutations are harmful or neutral — they can disrupt proteins and cause genetic diseases (for example sickle cell anaemia, phenylketonuria, haemophilia); they can cause cancers by damaging genes that control cell division; and in agriculture they can reduce yield or kill the organism. Harmful mutations are usually removed from populations by natural

selection because they reduce fitness.

Q12. How does sickle cell anaemia occur?

Ans. Sickle cell anaemia occurs due to a point mutation in the gene for the beta chain of haemoglobin: the codon GAG (glutamic acid) changes to GUG (valine) at the sixth position of the beta chain. The abnormal haemoglobin (HbS) makes red blood cells collapse into a sickle shape when oxygen is low; the sickled cells block small blood vessels, break down easily (causing anaemia) and cause pain and organ damage.

Q13. Which type of treatment is suitable for phenylketonuria?

Ans. Phenylketonuria (PKU) is treated by a strict diet low in phenylalanine, started as early as possible after birth: the child avoids protein-rich foods (meat, milk, eggs) and uses special phenylalanine-free formulas, because the body cannot convert phenylalanine to tyrosine and its accumulation damages the brain. The diet must usually be continued for life, and blood phenylalanine levels are monitored regularly.

Q14. Write symptoms of Turner's syndrome.

Ans. Symptoms of Turner's syndrome (45, XO — a female with only one X chromosome) are: short stature; webbed neck; underdeveloped ovaries (so no menstruation and infertility); broad chest with widely spaced nipples; swelling of hands and feet at birth; and sometimes heart and kidney defects and learning difficulties.

Q15. Differentiate between point mutation and frameshift mutation.

Ans. A point mutation is a change in a single base pair of DNA — a substitution, which may change one amino acid (missense), create a stop codon (nonsense) or have no effect (silent). A frameshift mutation is an insertion or deletion of one or two bases (not a multiple of three), which shifts the entire reading frame after that point, so every codon downstream is changed and usually a non-functional protein results. Point mutations affect one codon; frameshift mutations affect all codons after the mutation.

Q16. Differentiate between R-type and S-type Streptococcus.

Ans. In Griffith's experiment, R-type (rough, non-virulent) Streptococcus pneumoniae are harmless and S-type (smooth, virulent) kill mice. Heat-killed S-type bacteria alone do not kill mice, and living R-type alone do not kill mice — but a mixture of heat-killed S-type and living R-type kills mice, and living S-type bacteria are recovered from the dead mice. Griffith concluded that a 'transforming principle' from the dead S-type had changed the R-type into virulent S-type; Avery later showed this principle was DNA.

Q17. How can Turner's syndrome be treated?

Ans. Turner's syndrome cannot be cured, but it is treated: growth hormone injections in childhood increase height; oestrogen replacement therapy at puberty develops secondary sexual characters and maintains bone health; and associated problems (heart defects, kidney problems, hearing loss) are treated by specialists. Fertility treatment is usually not possible because the

ovaries do not develop.

Q18. Difference between primary constriction and secondary constriction.

Ans. The primary constriction is the centromere — the point where the two sister chromatids are joined and where spindle fibres attach during division; it is present in every chromosome and divides it into arms. The secondary constriction is an additional narrow region elsewhere on the chromosome (other than the centromere), often the site of the nucleolar organiser (where rRNA genes are located); it does not take part in spindle attachment.

Exercise - Section III: Extended Response Questions

Q1. Describe in detail the composition of chromosome.

Ans. A chromosome is composed of DNA and proteins. The DNA is a long double helix wound around histone proteins: eight histone molecules form a nucleosome (bead), with about 200 base pairs of DNA wrapped around it, and the nucleosomes are coiled into a 30-nm chromatin fibre, which loops and folds further into the compact chromosome seen during division. Each chromosome has two sister chromatids joined at the centromere (primary constriction), with a short and a long arm; the ends are protected by telomeres, and some chromosomes have a secondary constriction (nucleolar organiser). Non-histone proteins also help organise the structure.

Q2. Illustrate the experiment performed by Hershey-Chase.

Ans. Hershey and Chase (1952) proved DNA is the hereditary

material using bacteriophage T2. They grew one batch of phages in radioactive sulphur (35S), which labels protein only, and another in radioactive phosphorus (32P), which labels DNA only. They let the phages infect bacteria, then shook the mixture in a blender to remove the phage coats and centrifuged it. The 32P (DNA) was found inside the bacterial cells, while the 35S (protein) stayed outside in the coats. Since only DNA entered the bacteria and new phages were still produced, they concluded that DNA, not protein, carries hereditary information.

Q3. Describe elongation phase of DNA replication.

Ans. Elongation of DNA replication: after initiation at the origin, helicase unwinds the helix and single-strand binding proteins keep the strands apart, forming the replication fork. Primase lays down an RNA primer, and DNA polymerase III adds nucleotides to the 3' end, synthesising the leading strand continuously towards the fork and the lagging strand discontinuously away from the fork in Okazaki fragments. DNA polymerase I removes the RNA primers and fills the gaps with DNA, and DNA ligase seals the fragments into a continuous strand. Replication proceeds in both directions from each origin, giving two daughter molecules.

Q4. Describe gene mutation with its types.

Ans. A gene mutation is a permanent change in the base sequence of a gene. Its types are: point mutations — substitution of one base (missense, nonsense or silent); frameshift mutations — insertion or deletion of bases that shift the reading frame; and chromosomal mutations — changes in chromosome number (aneuploidy, for

example Down's, Turner's and Klinefelter's syndromes) or structure (deletion, duplication, inversion, translocation). Mutations are caused by mutagens (radiation and chemicals) or by errors in copying, and they may be harmful, neutral or, rarely, useful.

Q5. Describe the experiment of Frederic Griffith.

Ans. Griffith (1928) worked with Streptococcus pneumoniae in mice. Living smooth (S-type, virulent) bacteria killed mice; living rough (R-type, non-virulent) did not; heat-killed S-type did not; but a mixture of heat-killed S-type and living R-type killed mice, and living S-type bacteria were recovered from them. Griffith concluded that something from the dead S-type — the 'transforming principle' — had transformed the harmless R-type into virulent S-type. Avery, MacLeod and McCarty (1944) later identified the transforming principle as DNA.

Q6. Explain gene and genetic code.

Ans. A gene is a sequence of DNA that carries the information for making a functional product — a protein or an RNA molecule. The genetic code is the set of rules by which the base sequence of DNA (copied into mRNA) is translated into amino acids: it is read in triplets called codons, each codon coding for one amino acid; there are 64 codons for 20 amino acids, so the code is degenerate; it is universal, specific and non-overlapping, with AUG as the start codon and UAA, UAG and UGA as stop codons.

Q7. Describe post-transcriptional modification of mRNA.

Ans. Post-transcriptional modification of mRNA in eukaryotes: (1)

Capping — a modified guanine (7-methylguanosine cap) is added to the 5' end, protecting the mRNA and helping ribosome binding. (2) Tailing — a poly-A tail (100-200 adenine nucleotides) is added to the 3' end, protecting the mRNA from breakdown. (3) Splicing — introns (non-coding parts) are removed and exons (coding parts) are joined together by the spliceosome, forming the mature mRNA that leaves the nucleus for translation.

Q8. Explain how prokaryotic and eukaryotic protein synthesis differ from each other.

Ans. Prokaryotic protein synthesis: transcription and translation occur together in the cytoplasm (no nucleus), mRNA is usually polycistronic (carrying several genes), ribosomes are 70S (50S + 30S), and translation starts with formyl-methionine; there is no splicing or capping. Eukaryotic protein synthesis: transcription occurs in the nucleus and translation in the cytoplasm, mRNA is monocistronic and is capped, tailed and spliced before export, ribosomes are 80S (60S + 40S), and translation starts with methionine. Eukaryotic synthesis is therefore slower and more heavily processed.

Q9. Write a detailed description of Down's syndrome.

Ans. Down's syndrome (trisomy 21, 2n+1) is caused by non-disjunction of chromosome 21 during meiosis, so the child has three copies of chromosome 21 (47 chromosomes in total). Its features are: a flat facial profile, slanting eyes with an epicanthal fold, a short neck, short stature, a single palmar crease, poor muscle tone (hypotonia) and congenital heart defects. Intellectual

disability ranges from mild to moderate. There is no cure; management includes early education, speech and physiotherapy, treatment of heart defects, and family support.

Science, Technology and Society (STS) (model answers)

Q1. Paradoxical nature of DNA, as a tool of geneticists and forensics.

Ans. DNA is paradoxical as a tool: almost every cell contains the same DNA, and any two people share 99.9% of it, yet the remaining 0.1% (about 3 million base differences) is enough to identify a person uniquely. Forensic scientists scan 13 variable regions (loci) to build a DNA profile (fingerprint), used to identify suspects from crime-scene samples (hair, blood, saliva), to exonerate the wrongly accused, to identify disaster victims, to establish paternity and family relationships, and to study evolutionary relationships.

Q2. Explain how harmful mutations have been eradicated by nature.

Ans. Harmful mutations are removed by natural selection: a mutation that reduces an organism's fitness makes it less likely to survive to reproduce, or to leave as many offspring, so the harmful allele becomes rarer in each generation. Over many generations, harmful traits are eradicated from the population, while favourable traits spread. Recessive harmful alleles can persist hidden in carriers, which is why they are reduced but not always completely eliminated.

Chapter 13: Evolution

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. Endosymbiont hypothesis explains origin of:

- ✓ Eukaryotes
- Prokaryotes
- Armadillo
- Fish

Q2. The glyptodont mammal that lives only in America is:

- Echidna
- Pangolin
- Porcupine
- ✓ Armadillo

Q3. Who proposed endosymbiotic theory?

- Lamarck
- ✓ Margulis
- Darwin
- Mendel

Q4. All are Darwin's observations about natural selection except:

- over production
- struggle for existence
- ✓ use and disuse of organ
- variation

Q5. Analogous structures are:

- Anatomically similar and functioning similarly
- Anatomically similar but functioning differently
- ✓ Anatomically different but functioning similarly
- Anatomically different and functioning differently

Q6. The theory of inheritance of acquired characters was given by:

- Darwin
- ✓ Lamarck
- Aristotle
- Mendel

Q7. The descent with modification and natural selection are two main points of:

- Lamarckism
- Endosymbiotic theory
- Mutation theory
- ✓ Darwinism

Q8. Lamarck's theory of natural selection is usually known as:

- Natural selection
- ✓ Inheritance of acquired characters
- Descent with change
- Mutation

Q9. Biogeography is the geographical distribution of:

- phylum

✓ species

classes

genera

Q10. Who figured out the mechanism of evolution known as natural selection?

✓ Charles Darwin

Alfred Russell

Lamarck

Buffen

Q11. Who wrote the book 'Philosophie Zoologique'?

Linnaeus

Darwin

✓ Lamarck

Mendel

Q12. Respiratory protein found in all aerobic organisms:

cytochrome a

cytochrome b

✓ cytochrome c

cytochrome d

Q13. The force that initiates evolution is:

variations

extinction

adaptation

✓ mutation

Q14. Lamarck was an invertebrate zoologist and well known as:

- ✓ Toponymist
- Physiologist
- Biochemist
- Botanist

Exercise - Section II: Short Response Questions

Q1. How is convergent evolution related to analogous structure?

Ans. Convergent evolution is related to analogous structures because when unrelated organisms live in similar environments, natural selection shapes them in similar ways, so they develop structures that look alike and perform the same function although they have different origins — for example the wings of birds and insects, or the streamlined bodies of sharks and dolphins. Such similar structures in unrelated organisms are called analogous structures, and they are evidence of convergent evolution.

Q2. What is divergent evolution?

Ans. Divergent evolution is the process by which two or more related species, descended from a common ancestor, become more and more different over time as they adapt to different environments or ways of life. For example, the forelimbs of mammals (human arm, bat wing, whale flipper, horse leg) share the same basic bone plan but have diverged to perform very different functions.

Q3. How do analogous organs differ from homologous organs?

Ans. Analogous organs have the same function but different structure and origin (for example wings of birds and insects, eyes of octopus and humans). Homologous organs have the same basic structure and origin (common ancestry) but may perform different functions (for example the forelimbs of humans, bats, whales and cats). Analogous organs show convergent evolution; homologous organs show divergent evolution.

Q4. What is meant by descent with modification?

Ans. Descent with modification means that all species are descended from pre-existing species, with changes (modifications) accumulating in successive generations. According to Darwin, all species descended from only a few original life forms, and the small differences that appear in each generation, when favoured by natural selection, gradually produce new species. It is the backbone of Darwin's theory of evolution.

Q5. Who proposed endosymbiotic theory?

Ans. The endosymbiotic theory was proposed by Lynn Margulis (in the 1960s, building on earlier ideas). It states that mitochondria and chloroplasts originated as free-living prokaryotes that were engulfed by a larger host cell and began living inside it in a mutually beneficial relationship, eventually becoming permanent organelles.

Q6. What is meant by use and disuse of organs? Write example.

Ans. Use and disuse of organs is Lamarck's idea that an organ used often becomes larger and stronger, while an organ not used

becomes smaller and weaker, and these changes are inherited. Example: Lamarck argued that giraffes stretched their necks to reach high leaves, so their necks became longer over generations; and that snakes lost their limbs because they stopped using them.

Q7. What is meant by overproduction?

Ans. Overproduction means that every species produces far more offspring than can possibly survive: for example, a salmon lays more than 2.5 million eggs in one season, and a grasshopper lays more than 200 eggs at a time. If all survived, a single pair of elephants would give rise to 19 million descendants in 750 years. Since resources (food, space) are limited, most offspring die, which leads to the struggle for existence.

Q8. How does biogeography favour evolution as evidence?

Ans. Biogeography favours evolution because closely related species are found in the same geographical region, while different regions have different, unique species. For example, armadillos live in North and South America, where the fossil glyptodonts (their extinct relatives) also lived; and the unique marsupials of Australia suggest they evolved there in isolation from a common ancestor. This distribution is best explained by common descent and local evolution.

Q9. How does Giraffe neck support Lamarck's theory?

Ans. According to Lamarck, giraffes originally had short necks, but they stretched their necks to reach leaves on tall trees; the stretching made the neck longer during their lifetime, and this

acquired longer neck was inherited by their offspring. Over many generations of stretching and inheriting, giraffes developed the long necks seen today. (Modern biology does not accept this explanation: long necks evolved by natural selection favouring giraffes born with longer necks.)

Q10. How does limb loss in snake support Lamarck's theory?

Ans. According to Lamarck, snakes once had limbs, but as they adopted a burrowing and crawling way of life they stopped using their limbs; the unused limbs became smaller in each generation and were eventually lost, and this loss was inherited. So the limbless condition of snakes was, in Lamarck's view, the result of disuse of organs over many generations. (Modern biology explains limb loss by natural selection, not by disuse.)

Q11. Write two ideas of Lamarck's theory.

Ans. The two ideas of Lamarck's theory are: (1) Use and disuse of organs — organs used frequently become developed, while unused organs degenerate. (2) Inheritance of acquired characters — the changes acquired by an organism during its lifetime are passed on to its offspring. Together these ideas were meant to explain how species change over time.

Q12. How does convergent evolution differ from divergent evolution?

Ans. Convergent evolution is the development of similar features in unrelated organisms living in similar environments (for example wings of birds and insects), producing analogous structures.

Divergent evolution is the development of different features in related organisms descended from a common ancestor as they adapt to different environments (for example forelimbs of mammals), producing homologous structures. So convergence brings unrelated forms together, while divergence drives related forms apart.

Exercise - Section III: Extended Response Questions

Q1. Describe Lamarck's theory of evolution.

Ans. Lamarck's theory of evolution (1809, in 'Philosophie Zoologique') rests on two ideas. (1) Use and disuse: organs that are used become stronger and larger, while organs that are not used weaken and disappear. (2) Inheritance of acquired characters: changes acquired during an organism's lifetime are inherited by its offspring. Examples: giraffes lengthened their necks by stretching for leaves, and snakes lost their limbs by disuse. The theory was disproved by August Weismann, who cut off the tails of mice for many generations yet the offspring were always born with tails, showing that acquired characters are not inherited.

Q2. Explain Darwin's theory of natural selection.

Ans. Darwin's theory of natural selection (1859) has two main points: descent with modification and natural selection. Its four observations are: (1) Overproduction — species produce more offspring than can survive. (2) Struggle for existence — individuals compete for limited food and space (intraspecific, interspecific and environmental struggle). (3) Variations — individuals differ in heritable traits, arising from mutation and recombination. (4)

Survival of the fittest — individuals with favourable variations survive and reproduce more, so favourable alleles increase and unfavourable ones decrease; over many generations this produces new species adapted to their environment.

Q3. Describe evidences in favour of evolution.

Ans. Evidences in favour of evolution: (1) Fossil record — fossils show simpler organisms in older rocks and more complex ones in younger rocks, and transitional forms (for example Archaeopteryx between reptiles and birds). (2) Comparative anatomy — homologous structures (mammalian forelimbs) show common ancestry, while analogous structures show convergent evolution. (3) Embryology — early embryos of vertebrates look very similar. (4) Biogeography — related species occur in the same region (armadillos and fossil glyptodonts in the Americas). (5) Molecular evidence — all organisms share the same genetic code and proteins such as cytochrome c, with differences proportional to evolutionary distance. (6) Vestigial organs — for example the human appendix and the pelvic bones of whales.

Q4. Explain evolution of eukaryotes from prokaryotes through endosymbiotic theory.

Ans. The endosymbiotic theory (Margulis) explains the evolution of eukaryotes from prokaryotes: about 2 billion years ago, a large anaerobic prokaryote engulfed smaller aerobic bacteria, which survived inside it instead of being digested and became mitochondria; later, a similar engulfment of a photosynthetic cyanobacterium gave rise to chloroplasts. Evidence: mitochondria

and chloroplasts have their own circular DNA, their own 70S ribosomes, a double membrane, and they divide by fission like bacteria. Thus the eukaryotic cell is a combination of once-independent prokaryotes.

Q5. Write a note on overproduction and survival of the fittest.

Ans. Overproduction: every species produces far more offspring than the environment can support — a salmon lays millions of eggs, an elephant pair could produce 19 million descendants in 750 years. Because food and space are limited, a struggle for existence follows (within a species, between species, and against the environment). Survival of the fittest: individuals with variations better suited to the environment survive and reproduce more successfully, passing their favourable traits to the next generation, while less fit individuals are eliminated. Over generations, the population becomes better adapted, and new species may arise.

Chapter 14: Ecology

Exercise - Section I: Multiple Choice Questions (MCQs)

Q1. One of the following is not an interaction found within a community.

- Predation
- Parasitism
- ✓ Photosynthesis
- Symbiosis

Q2. The heterotroph which obtains food from living hosts is:

- Man
- ✓ Cuscuta
- Green plants
- Fungi

Q3. Ammonia is converted into nitrite (NO_2) by:

- Nitrobacter bacteria
- Pseudomonas bacteria
- ✓ Nitrosomonas bacteria
- Azotobacter

Q4. Which of these levels of ecological study involves both abiotic and biotic components?

- Organisms
- Population

✓ Ecosystem

Community

Q5. Change in community structure of an ecosystem over a period of time is called:

Niche

Unstable ecosystem

✓ Succession

Pioneer

Q6. The main pollutants responsible for acid rain are:

Oxygen and Hydrogen

✓ Sulphur dioxide and nitric acid

Ozone and carbon dioxide

Carbon monoxide and methane

Q7. Herbaceous stage in xerosere is the:

First stage

Third Stage

✓ Fourth stage

Last stage

Q8. The group of organisms that fix atmospheric nitrogen are:

Plants

✓ Bacteria

Fungi

Insects

Q9. Coral reefs are called 'Rain forests of the sea' because:

They receive a lot of rain

✓ They provide habitat and shelter to many marine organisms

They are located in rain forest

They grow in rain water

Q10. The main cause of greenhouse effect is:

✓ CO_2

Hydrogen

Nitrogen

Oxygen

Q11. Tertiary consumers are classified as:

T5 level

T3 level

✓ T4 level

T2 level

Q12. The percentage of lost energy between one trophic level to the next is:

70%

85%

80%

✓ 90%

Q13. In Xerarch, the next stage after foliose lichen stage is:

Crustose lichens stage

Shrub stage

✓ Moss stage

Herbaceous stage

Q14. An aquifer is:

A basin or channel where water is stored

A type of river

A type of water cycle driven by solar energy

✓ An underground layer where the materials containing water

Q15. The total amount of solar energy fixed by producers during photosynthesis in a given area over a specific time is called:

Net primary production

✓ Gross primary production

Biomass

Respiration

Exercise - Section II: Short Response Questions

Q1. What do you know about the various modes of nutrition?

Ans. The modes of nutrition are: (1) Autotrophic nutrition — organisms make their own food: photoautotrophs (green plants, algae) use sunlight, and chemoautotrophs (nitrifying bacteria) use chemical energy. (2) Heterotrophic nutrition — organisms obtain food from others: herbivores eat plants, carnivores eat animals, omnivores eat both; saprophytes (fungi, bacteria) absorb food from dead matter; parasites (Cuscuta, tapeworm) obtain food from living hosts; and symbionts share food mutually (lichens).

Q2. Write a note on the loss of energy between trophic levels.

Ans. Energy is lost between trophic levels because: not all food is eaten; much of the food eaten is not digested and is lost as faeces; and most of the absorbed energy is used in respiration, movement and keeping warm, and is lost as heat. Only about 10% of the energy at one level is transferred to the next level (the 10% law), so food chains rarely have more than 4-5 levels and the top carnivores receive very little energy.

Q3. Write a short note on biomass pyramid.

Ans. A biomass pyramid is a diagram showing the total dry mass of organisms at each trophic level of a food chain, usually as horizontal bars. In most land ecosystems it is upright: producers have the largest biomass, herbivores less, and top carnivores the least, because energy is lost at each transfer. In some aquatic ecosystems it can be inverted, because the small, fast-reproducing phytoplankton have less standing biomass than the zooplankton that feed on them.

Q4. How will you classify the ecological succession on the basis of location?

Ans. On the basis of location, ecological succession is classified into: (1) Hydrarch (hydrosere) — succession starting in water (ponds, lakes), from plankton and submerged plants to a land climax community. (2) Xerarch (xerosere) — succession starting in dry places (bare rock, sand), from lichens and mosses to herbs, shrubs and finally a forest climax. (3) Mesarch — succession in places with moderate moisture. Primary succession starts on bare ground, while secondary succession starts where a community was

destroyed but soil remains.

Q5. Differentiate between industrial and biological fixation of nitrogen.

Ans. Industrial fixation of nitrogen is done by humans in factories (the Haber process): nitrogen and hydrogen are combined at high temperature and pressure to make ammonia for fertilisers; it needs fossil fuel energy and produces large amounts at once. Biological fixation is done by living organisms (Rhizobium in legume roots, Azotobacter and cyanobacteria in soil): the enzyme nitrogenase fixes nitrogen at normal temperature and pressure, slowly and continuously, as part of the natural nitrogen cycle.

Q6. Differentiate between Hydrarch and Xerarch.

Ans. Hydrarch succession begins in water: pioneer plankton and submerged plants colonise a pond, dead matter accumulates, the pond becomes shallower, marsh plants appear, then shrubs and finally a land climax forest. Xerarch succession begins on dry bare rock: crustose lichens are pioneers, followed by foliose lichens, mosses, herbs, shrubs and finally a forest climax. So hydrarch goes from wet to dry, while xerarch goes from dry to mesic, but both end in a stable climax community.

Q7. Write a short note on the human impacts on an ecosystem.

Ans. Human impacts on ecosystems include: deforestation for farming and timber, which destroys habitats; overgrazing, which makes land barren; pollution of air, water and soil by industry, vehicles and plastics; overfishing, especially in breeding seasons;

burning of fossil fuels, causing the greenhouse effect and global warming; and the introduction of exotic species. These reduce biodiversity, disturb food webs and degrade the resources humans depend on.

Q8. How are coral reefs important for a marine ecosystem?
Ans. Coral reefs are important for marine ecosystems because they provide habitat and shelter to a huge variety of marine organisms — fish, molluscs, crustaceans and algae — like rainforests on land, so they are called the 'rainforests of the sea'. They protect coastlines from waves and storms, support fisheries that feed millions of people, and are a source of medicines and tourism income.

Q9. Define the following terms: (i) Aquifer (ii) Net primary productivity (iii) Climax community (iv) Biogeochemical Cycle.
Ans. (i) Aquifer: an underground layer of permeable rock or sediment (sand, gravel) that holds groundwater and can supply wells and springs. (ii) Net primary productivity: the rate at which producers store energy as new biomass after subtracting what they use in respiration — that is, gross primary productivity minus respiration; it is the food actually available to herbivores. (iii) Climax community: the final, stable community of a succession, in equilibrium with the climate, which does not change further unless disturbed. (iv) Biogeochemical cycle: the circulation of chemical elements and compounds (carbon, nitrogen, water) between living organisms and their physical environment.

Exercise - Section III: Extended Response Questions

Q1. What is ocean acidification? How does it affect marine ecosystem?

Ans. Ocean acidification is the process by which the ocean becomes more acidic because it absorbs excess carbon dioxide from the atmosphere. The CO2 dissolves in seawater and forms carbonic acid, which lowers the pH. This affects marine ecosystems seriously: it reduces the carbonate ions that corals, shellfish, oysters and plankton need to build their shells and skeletons, so coral reefs weaken and grow slowly; it harms the food chains that depend on these organisms and threatens fisheries and the livelihoods of coastal people.

Q2. Describe water cycle.

Ans. The water cycle: water evaporates from oceans, lakes and land and transpires from plants, rising as water vapour. The vapour cools and condenses into clouds, then falls as precipitation (rain, snow) on land and sea. On land, water runs off into rivers, infiltrates the soil to become groundwater (stored in aquifers) or is taken up by plants; rivers carry it back to the sea. The cycle is driven by solar energy and gravity, and it continuously purifies and redistributes water.

Q3. Write a comprehensive note on greenhouse effect.

Ans. The greenhouse effect is the warming of the Earth because certain gases (carbon dioxide, methane, water vapour, nitrous oxide) in the lower atmosphere absorb infrared radiation radiated by the Earth and re-radiate it back, acting like the glass of a

greenhouse. It is natural and keeps the Earth warm enough for life, but burning fossil fuels, deforestation and industry have increased CO2 and other gases, strengthening the effect. The result is global warming: melting glaciers and sea-level rise, changing rainfall, more extreme weather, and shifts in ecosystems and agriculture.

Q4. Explain the characteristics of population dynamics.

Ans. Population dynamics is the study of long-term and short-term changes in the size, age structure and composition of populations. Its characteristics are: populations grow when births and immigration exceed deaths and emigration; growth is limited by carrying capacity (the maximum population an environment can support); growth may be exponential when resources are abundant and logistic (S-shaped) when limited; and populations show fluctuations due to predation, disease, competition and environmental change. Age pyramids (expanding, stable, declining) predict future growth.

Q5. Explain different stages of xerosere succession.

Ans. Xerosere succession (on bare rock) has these stages: (1) Crustose lichen stage — crusty lichens are pioneers; they secrete acids that weather the rock and trap dust, forming a thin soil. (2) Foliose lichen stage — leafy lichens grow on the improved surface. (3) Moss stage — mosses hold more water and add humus. (4) Herbaceous stage — grasses and herbs appear. (5) Shrub stage — shrubs shade out herbs and deepen the soil. (6) Woodland/climax stage — trees establish, forming a stable forest community in equilibrium with the climate.

Q6. What are trophic levels? Explain different levels of trophic levels.

Ans. Trophic levels are the feeding positions in a food chain or food web. The levels are: T1 Producers — green plants and algae that make food by photosynthesis. T2 Primary consumers (herbivores) — animals that eat producers (for example deer, grasshoppers). T3 Secondary consumers (carnivores) — animals that eat herbivores (for example frogs, foxes). T4 Tertiary consumers — animals that eat secondary consumers (for example lions, eagles). T5 Apex predators — top carnivores with no natural enemies. Decomposers (bacteria, fungi) act at all levels, returning nutrients to the soil.

Q7. What are trophic levels? What do you know about the sequence of trophic levels? Explain each level with example.

Ans. Trophic levels are the steps of a food chain, each representing a feeding group. Their sequence, with examples, is: Producers (T1) — grass, algae, which trap solar energy. Primary consumers (T2) — grasshopper, deer, zooplankton, which eat producers. Secondary consumers (T3) — frog, fox, small fish, which eat herbivores. Tertiary consumers (T4) — snake, lion, large fish, which eat secondary consumers. Apex predators (T5) — eagle, tiger, which are eaten by none. At each transfer about 90% of energy is lost as heat and waste, so only about 10% passes on, which is why chains are short and top predators are few.

Q8. Write a detailed note on desert ecosystem in Pakistan.

Ans. Pakistan has two kinds of deserts. Warm deserts: Thal in

western Punjab (Mianwali, Bhakkar), Cholistan in southern Punjab (Fort Abbas, Bahawalnagar, Yazman, Bahawalpur, Khanpur, Rahim Yar Khan), Thar (in Sindh, bordering Cholistan) and Kharan in Balochistan. In summer temperatures exceed 40 degrees C and sometimes reach 50 degrees C; winters may fall below 10 degrees C; rainfall is only 100-250 mm per year, unreliable and monsoon-dependent. Cold desert: Katpana in Gilgit-Baltistan, with summer temperatures of 24-27 degrees C and winters from -17 degrees C to -25 degrees C, with snowfall. Vegetation is sparse and xerophytic (Acacia, cacti, Euphorbia, palms, grasses); animals are adapted to drought (camels, desert foxes, lizards, snakes, scorpions, bustards). Plants have deep or shallow roots, fleshy water-storing stems, waxy cuticles, sunken stomata and spiny leaves; animals are often nocturnal, live in burrows, lack sweat glands and excrete concentrated urine.

Q9. Explain nitrogen cycle in detail.

Ans. The nitrogen cycle: (1) Nitrogen fixation — atmospheric nitrogen is fixed into ammonia by nitrogen-fixing bacteria (Rhizobium in legume roots, Azotobacter in soil, cyanobacteria) and by lightning and industry. (2) Nitrification — Nitrosomonas bacteria convert ammonia into nitrites, and Nitrobacter convert nitrites into nitrates, which plants absorb. (3) Assimilation — plants use nitrates to make proteins and nucleic acids, which pass to animals through food. (4) Ammonification — decomposers convert dead matter and animal waste into ammonia. (5) Denitrification — Pseudomonas bacteria convert nitrates back into nitrogen gas, returning it to the atmosphere. The cycle keeps nitrogen available to living things.

Q10. Define acid rain. How does it occur? What are its effects on living and non-living things?

Ans. Acid rain is rain with a pH below 5.6, made acidic by pollutants. It occurs when sulphur dioxide (from burning coal and oil) and nitrogen oxides (from vehicles and industry) rise into the atmosphere and react with water vapour and oxygen to form sulphuric acid and nitric acid, which fall as rain, snow or fog. Its effects: on living things — it damages leaves and forests, acidifies lakes and kills fish and aquatic life, and harms human lungs and skin; on non-living things — it corrodes buildings, monuments (for example marble of the Taj Mahal) and metal structures, and leaches nutrients from soil, reducing fertility.

STEAM Activity (model answer)

Q1. Write down the amount of energy in each level of the energy pyramid (producers 100,000 J).

Ans. With 10% transfer at each level and producers at 100,000 J: Herbivores receive 10,000 J; Carnivores receive 1,000 J; Secondary carnivores receive 100 J; Apex predators receive 10 J. (The pyramid printed in the book labels producers as 10,000 J; using the book's printed 10,000 J base, the levels are Herbivores 1,000 J, Carnivores 100 J, Secondary carnivores 10 J, Apex predators 1 J.)